

Exam Questions

Section VI - Metabolism of Proteins & Amino Acids

- Select the one of the following statements that is NOT CORRECT:
 - Δ^1 -Pyrroline-5-carboxylate is an intermediate both in the biosynthesis and in the catabolism of L-proline.
 - Human tissues can form dietarily nonessential amino acids from amphibolic intermediates or from dietarily essential amino acids.
 - Human liver tissue can form serine from the glycolytic intermediate 3-phosphoglycerate.
 - The reaction catalyzed by phenylalanine hydroxylase interconverts phenylalanine and tyrosine.
 - The reducing power of tetrahydrobiopterin derives ultimately from NADPH.
- Identify the metabolite that does NOT serve as a precursor of a dietarily essential amino acid:
 - α -Ketoglutarate
 - 3-Phosphoglycerate
 - Glutamate
 - Aspartate
 - Histamine
- Select the one of the following statements that is NOT CORRECT:
 - Selenocysteine is present at the active sites of certain human enzymes.
 - Selenocysteine is inserted into proteins by a posttranslational process.
 - Transamination of dietary α -keto acids can replace the dietary essential amino acids leucine, isoleucine, and valine.
 - Conversion of peptidyl proline to peptidyl-4-hydroxyproline is accompanied by the incorporation of oxygen into succinate.
 - Serine and glycine are interconverted in a single reaction in which tetrahydrofolate derivatives participate.
- Select the CORRECT answer:

The first reaction in the degradation of most of the protein amino acids involves the participation of:

 - NAD⁺
 - Thiamine pyrophosphate (TPP)
 - Pyridoxal phosphate
 - FAD
 - NAD⁺ and TPP
- Identify the amino acid that is the major contributor to the transport of nitrogen destined for excretion as urea:
 - Alanine
 - Glutamine
 - Glycine
 - Lysine
 - Ornithine
- Select the one of the following statements that is NOT CORRECT:
 - Angelman syndrome is associated with a defective ubiquitin E3 ligase.
 - Following a protein-rich meal, the splanchnic tissues release predominantly branched-chain amino acids, which are taken up by peripheral muscle tissue.
 - The rate of hepatic gluconeogenesis from glutamine exceeds that of any other amino acid.
 - The L- α -amino oxidase-catalyzed conversion of an α -amino acid to its corresponding α -keto acid is accompanied by the release of NH₄⁺.
 - Similar or even identical signs and symptoms can be associated with different mutations of the gene that encodes a given enzyme.
- Select the one of the following statements that is NOT CORRECT:
 - PEST sequences target some proteins for rapid degradation.
 - ATP and ubiquitin typically participate in the degradation of membrane-associated proteins and other proteins with long half-lives.
 - Ubiquitin molecules are attached to target proteins via non- α peptide bonds.
 - The discoverers of ubiquitin-mediated protein degradation received a Nobel Prize.
 - Degradation of ubiquitin-tagged proteins takes place in the proteasome, a multi-subunit macromolecule present in all eukaryotes.
- For metabolic disorders of the urea cycle, which statement is NOT CORRECT:
 - Ammonia intoxication is most severe when the metabolic block in the urea cycle occurs prior to the reaction catalyzed by argininosuccinate synthase.
 - Clinical symptoms include mental retardation and the avoidance of protein-rich foods.
 - Clinical signs can include acidosis.
 - Aspartate provides the second nitrogen of argininosuccinate.
 - Dietary management focuses on a low-protein diet ingested as frequent small meals.
- Select the one of the following statements that is NOT CORRECT:
 - One metabolic function of glutamine is to sequester nitrogen in a nontoxic form.
 - Liver glutamate dehydrogenase is allosterically inhibited by ATP and activated by ADP.
 - Urea is formed both from absorbed ammonia produced by enteric bacteria and from ammonia generated by tissue metabolic activity.
 - The concerted action of glutamate dehydrogenase and glutamate aminotransferase may be termed transdeamination.
 - Fumarate generated during biosynthesis of argininosuccinate ultimately forms oxaloacetate in reactions in mitochondria catalyzed successively by fumarase and malate dehydrogenase.

10. Select the one of the following statements that is NOT CORRECT:
- Threonine provides the thioethanol moiety for biosynthesis of coenzyme A.
 - Histamine arises by decarboxylation of histidine.
 - Ornithine serves as a precursor of both spermine and spermidine.
 - Serotonin and melatonin are metabolites of tryptophan.
 - Glycine, arginine, and methionine each contribute atoms for biosynthesis of creatine.
11. Select the one of the following statements that is NOT CORRECT:
- Excreted creatinine is a function of muscle mass, and can be used to determine whether a patient has provided a complete 24-hour urine specimen.
 - Many drugs and drug catabolites are excreted in urine as glycine conjugates.
 - The major nonprotein metabolic fate of methionine is conversion to S-adenosylmethionine.
 - The concentration of histamine in brain hypothalamus exhibits a circadian rhythm.
 - Decarboxylation of glutamine forms the inhibitory neurotransmitter GABA (γ -aminobutyrate).
12. What distinguishes the routes by which each of the following amino acids appears in human proteins?
- 5-Hydroxylysine
 γ -Carboxyglutamate
Selenocysteine
13. What evolutionary advantage might be gained by the fact that certain amino acids are *dietarily* essential for human subjects?
14. What explanation can you offer to explain that metabolic defects that result in the complete absence of the activity of glutamate dehydrogenase have not been detected?
15. Which of the following is NOT a hemoprotein?
- Myoglobin
 - Cytochrome *c*
 - Catalase
 - Cytochrome P450
 - Albumin
16. A 30-year-old man presented at clinic with a history of intermittent abdominal pain and episodes of confusion and psychiatric problems. Laboratory tests revealed increases of urinary δ -aminolevulinate and porphobilinogen. Mutational analysis revealed a mutation in the gene for uroporphyrinogen I synthase (porphobilinogen deaminase). The probable diagnosis was:
- Acute intermittent porphyria.
 - X-linked sideroblastic anemia.
 - Congenital erythropoietic porphyria.
 - Porphyria cutanea tarda.
 - Variegate porphyria.
17. Select the one of the following statements that is NOT CORRECT:
- Bilirubin is a cyclic tetrapyrrole.
 - Albumin-bound bilirubin is transported to the liver.
 - High levels of bilirubin can cause damage to the brains of newborn infants.
 - Bilirubin contains methyl and vinyl groups.
 - Bilirubin does not contain iron.
18. A 62-year-old female presented at clinic with intense jaundice, steadily increasing over the preceding 3 months. She gave a history of severe upper abdominal pain, radiating to the back, and had lost considerable weight. She had noted that her stools had been very pale for some time. Lab tests revealed a very high level of direct bilirubin, and also elevated urinary bilirubin. The plasma level of alanine aminotransferase (ALT) was only slightly elevated, whereas the level of alkaline phosphatase was markedly elevated. Abdominal ultrasonography revealed no evidence of gallstones. Of the following, which is the most likely diagnosis?
- Gilbert syndrome
 - Hemolytic anemia
 - Type 1 Crigler-Najjar syndrome
 - Carcinoma of the pancreas
 - Infectious hepatitis
19. Clinical laboratories typically use diazotized sulfanilic acid to measure serum bilirubin and its derivatives. What is the physical basis that permits the laboratory to report results to the physician in terms of these two forms of bilirubin?
20. What signals the synthesis of heme to take place?

Exam Questions

Section VII – Structure, Function, & Replication of Informational Macromolecules

- Which of the following statements about β,γ -methylene and β,γ -imino derivatives of purine and pyrimidine triphosphates is CORRECT?
 - They are potential anticancer drugs.
 - They are precursors of B vitamins.
 - They readily undergo hydrolytic removal of the terminal phosphate.
 - They can be used to implicate involvement of nucleotide triphosphates by effects other than phosphoryl transfer.
 - They serve as polynucleotide precursors.
- Which of the following statements about nucleotide structures is NOT CORRECT?
 - Nucleotides are polyfunctional acids.
 - Caffeine and theobromine differ structurally solely with respect to the number of methyl groups attached to their ring nitrogens.
 - A purine is a heterocyclic aromatic molecule composed of a pyrimidine ring fused to an imidazole ring.
 - NAD⁺, FMN, S-adenosylmethionine, and coenzyme A all are derivatives of ribonucleotides.
 - 3',5'-Cyclic AMP and 3',5'-cyclic GMP (cAMP and cGMP) serve as second messengers in human physiology.
- Which of the following statements about purine nucleotide metabolism is NOT CORRECT?
 - An early step in purine biosynthesis is the formation of PRPP (phosphoribosyl 1-pyrophosphate).
 - Inosine monophosphate (IMP) is a precursor of both AMP and GMP.
 - Orotic acid is an intermediate in pyrimidine nucleotide biosynthesis.
 - Humans catabolize uridine and pseudouridine by analogous reactions.
 - Ribonucleotide reductase converts nucleoside diphosphates to the corresponding deoxyribonucleoside diphosphates.
- Which of the following statements is NOT CORRECT?
 - Metabolic disorders are only infrequently associated with defects in the catabolism of purines.
 - Immune dysfunctions are associated both with a defective adenosine deaminase and with a defective purine nucleoside phosphorylase.
 - The Lesch-Nyhan syndrome reflects a defect in hypoxanthine-guanine phosphoribosyl transferase.
 - Xanthine lithiasis can be due to a severe defect in xanthine oxidase.
 - Hyperuricemia can result from conditions such as cancer characterized by enhanced tissue turnover.
- Which of the following components are found in DNA? Choose the most complete answer.
 - A phosphate group, adenine, and ribose
 - A phosphate group, guanine, and deoxyribose
 - Cytosine and ribose
 - Thymine and deoxyribose
 - A phosphate group and adenine
- The backbone of a DNA molecule consists of which of the following?
 - Alternating sugars and nitrogenous bases
 - Nitrogenous bases alone
 - Phosphate groups alone
 - Alternating phosphate and sugar groups
 - Five carbon sugars alone
- Which of the following is the interconnecting bond that connects the nucleotides of RNA and DNA?
 - N-glycosidic bonds
 - 3'-5' phosphodiester linkages
 - Phosphomonoesters
 - 2' phosphodiester linkages
 - Peptide nucleic acid bonds
- Which component of the DNA duplex causes the molecule to have a net negative charge at physiologic pH?
 - Deoxyribose
 - Ribose
 - Phosphate groups
 - Chlorine ion
 - Adenine
- Which molecular feature listed causes duplex DNA to exhibit a near-constant width along its long axis?
 - A purine nitrogenous base always pairs with another purine nitrogenous base.
 - A pyrimidine nitrogenous base always pairs with another pyrimidine nitrogenous base.
 - A pyrimidine nitrogenous base always pairs with a purine nitrogenous base.
 - Repulsion between phosphate groups keeps the strands a uniform distance apart.
 - Attraction between phosphate groups keeps the strands a uniform distance apart.
- The model for DNA replication first proposed by Watson and Crick posited that every newly replicated double-stranded daughter duplex DNA molecule.
 - Was composed of the two strands from the parent DNA molecule
 - Contained solely the two newly synthesized strands of DNA
 - Contained two strands that are random mixtures of new and old DNA within each strand
 - Was composed of one strand derived from the original parental DNA duplex and one strand that was newly synthesized
 - Was composed of nucleotide sequences completely distinct from either parental DNA strand

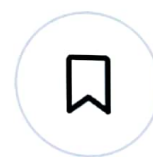
11. Name the mechanism through which RNAs are synthesized from DNA.
 - A. Replicational duplication
 - B. Translation
 - C. Translesion repair
 - D. Transesterification
 - E. Transcription
12. Which of the forces or interactions listed below play the predominant role in driving RNA secondary and tertiary structure formation?
 - A. Hydrophilic repulsion
 - B. Formation of complementary base pair regions
 - C. Hydrophobic interaction
 - D. van der Waals interactions
 - E. Salt bridge formation
13. Name the enzyme that synthesizes RNA from a double-stranded DNA template.
 - A. RNA-dependent RNA polymerase
 - B. DNA-dependent RNA convertase
 - C. RNA-dependent replicase
 - D. DNA-dependent RNA polymerase
 - E. Reverse transcriptase
14. Define the most notable characteristic difference with regard to gene expression between eukaryotes and prokaryotes.
 - A. Ribosomal RNA nucleotide lengths
 - B. Mitochondria
 - C. Lysosomes and peroxisomes
 - D. Sequestration of the genomic material in the nucleus
 - E. Chlorophyll
15. Which entry below correctly describes the approximate number of bp of DNA_____, which is separated into _____ chromosomes in a typical diploid human cell in a nonreplicating state?
 - A. 64 billion, 23
 - B. 6.4 trillion, 46
 - C. 23 billion, 64
 - D. 64 billion, 46
 - E. 6.4 billion, 46
16. What is the approximate number of base pairs associated with a single nucleosome?
 - A. 146
 - B. 292
 - C. 73
 - D. 1460
 - E. 900
17. All but one of the following histones are found located within the superhelix formed between DNA and the histone octamer; which of the following is this histone?
 - A. Histone H2B
 - B. Histone H3
 - C. Histone H1
 - D. Histone H3
 - E. Histone H4
18. Chromatin can be broadly defined as active and repressed. Which of the following is termed as a subclass of chromatin that is specifically inactivated at certain times within an organism's life and/or in particular sets of differentiated cells?
 - A. Constitutive euchromatin
 - B. Facultative heterochromatin
 - C. Euchromatin
 - D. Constitutive heterochromatin
19. Which of the following hypothesizes that the physical and functional status of a certain region of genomic chromatin is dependent on the patterns of specific histone posttranslational modifications (PTMs), and/or DNA methylation status?
 - A. Morse code
 - B. PTM hypothesis
 - C. Nuclear body hypothesis
 - D. Epigenetic code
 - E. Genetic code
20. What is the name of the unusual repeated stretch of DNA localized at the tips of all eukaryotic chromosomes?
 - A. Kinetochore
 - B. Telomere
 - C. Centriole
 - D. Chromomere
 - E. Micromere
21. Given that DNA polymerases are unable to synthesize DNA without a primer, what molecule serves as the primer for these enzymes during DNA replication?
 - A. Five carbon sugars
 - B. Deoxyribose alone
 - C. A short RNA molecule
 - D. Proteins with free hydroxyl groups
 - E. Phosphomonoester
22. Which of the following terms is used for the discontinuous DNA replication that occurs during replication is catalyzed via the production of small DNA segments?
 - A. Okazaki fragments
 - B. Toshihiro pieces
 - C. Onishi oligonucleotides
 - D. Crick strands
 - E. Watson fragments
23. What molecule or force supplies the energy that drives the relief of mechanical strain by DNA gyrase?
 - A. Pyrimidine to purine conversion
 - B. Hydrolysis of GTP
 - C. Hydrolysis of ATP
 - D. Glycolysis
 - E. A proton gradient molecule or force
24. What is the name of the phase of the cell cycle between the conclusion of cell division and the beginning of DNA synthesis?
 - A. G₁
 - B. S
 - C. G₂
 - D. M
 - E. G₀

25. At what stage of the cell cycle are key protein kinases, like cyclin-dependent kinase, activated?
- Right before mitosis
 - At the beginning of S phase
 - Near the end of G₁ phase
 - At the end of the G₂ phase
 - All of the above
26. What disease is often associated with a breakdown of a cell's ability to regulate/control its own division?
- Kidney disease
 - Cancer
 - Emphysema
 - Diabetes
 - Heart disease
27. What is the molecular mechanism that is responsible for the quick decrease in the Cdk activity that leads to exit from the M phase and the entry into G₁?
- Drop in mitotic cyclin concentration
 - Decreased G₁ cyclin concentration
 - Rise in G₂ cyclin concentration
 - Rise in mitotic cyclin concentration
 - Rise in G₁ cyclin concentration
28. Which of the following is the site to which RNA polymerase binds on the DNA template prior to the initiation of transcription?
- Intron/exon junction
 - Open reading frame DNA
 - Terminator
 - Initiator methionine codon
 - Promoter
29. The large eukaryotic rRNA genes, such as 18S and 28S RNA-encoding genes, are transcribed by which of the following RNA polymerases?
- RNA polymerase III
 - RNA-dependent RNA polymerase δ
 - RNA polymerase I
 - RNA polymerase II
 - Mitochondrial RNA polymerase
30. Eukaryotic RNA polymerases all have a requirement for a large variety of accessory proteins to enable them to bind promoters and form physiologically relevant transcription complexes. What are these proteins termed as?
- Basal or general transcription factors
 - Activators
 - Accessory factors
 - Elongation factors
 - Facilitator polypeptides
31. The DNA segment from which the primary transcript is copied or transcribed is called which of the following?
- Coding strand
 - Initiator methionine domain
 - Translation unit
 - Transcriptome
 - Initial codon
32. What class of DNA sequences are the eukaryotic genes that encode rRNAs?
- Single-copy DNA
 - Highly repetitive DNA
 - Moderately repetitive DNA
 - Mixed sequence DNA
33. How the modifications to the nucleotides of the pre-tRNAs, pre-rRNAs, and pre-mRNAs occur?
- Postprandially
 - Postmitotically
 - Pretranscriptionally
 - Posttranscriptionally
 - Prematurely
34. RNA polymerase II promoters are located on which side of the transcription unit?
- Internal
 - 3' downstream
 - Nearest to the C-terminus
 - Nearest to the N-terminus
 - 5' upstream
35. With regard to eukaryotic mRNAs, which one of the following is not a normal property of mRNAs?
- Eukaryotic mRNAs have special modifications at their 5' (cap) and 3' (poly(A) tail) termini.
 - Are attached to ribosomes when they are translated.
 - They are found in the cytoplasm within peroxisomes.
 - Most have a significant noncoding segment that does not direct assembly of amino acids.
 - Contain continuous nucleotide sequences that encode a particular polypeptide.
36. Which of the following is the bond connecting the initiation nucleotide of the mRNA with the 5^{me}-G Cap structure?
- 3'-5' phosphodiester bridge
 - 5'-5' triphosphate bridge
 - 3'-3' triphosphate bridge
 - 3'-5' triphosphate bridge
 - 5'-3' triphosphate bridge
37. What sequence feature of mature mRNAs listed below is thought to protect mRNAs from degradation?
- Special posttranslational modifications
 - 3' Poly(C)_n tail
 - 5^{me}-G Cap
 - Introns
 - Lariat structures
38. What could the consequences of inaccurate mRNA splicing be for the RNA?
- A single base error at a splice junction will cause a large deletion.
 - A single base error at a splice junction will cause a large insertion.
 - A single base error at a splice junction will cause a large inversion.
 - C and E.
 - A single base error at a splice junction will change the reading frame and result in mRNA mistranslation.

39. What is the macromolecular complex that associates with introns during mRNA splicing?
- Splicer
 - Dicer
 - Nuclear body
 - Spliceosome
 - Slicer
40. What reaction does reverse transcriptase catalyze?
- Translation of RNA to DNA
 - Transcription of DNA to RNA
 - Conversion of ribonucleotides into deoxyribonucleotides
 - Transcription of RNA to DNA
 - Conversion of a ribonucleotide to deoxynucleotides in the DNA double helix
41. RNAi or dsRNA-mediated RNA interference mediates which of the following?
- RNA ligation
 - RNA silencing
 - RNA inversion
 - RNA restoration
 - RNA quelling
42. While the genetic code has 64 codons, there are only 20 naturally occurring amino acids. Consequently, some amino acids are encoded by more than one codon. This feature of the genetic code is an illustration of the genetic code being which of the following?
- Degenerative
 - Duplicative
 - Nonoverlapping
 - Overlapping
 - Redundant
43. The genetic code contains how many termination codons?
- 3
 - 21
 - 61
 - 64
 - 20
44. If a tRNA has the sequence 5'-CAU-3', what codon would it recognize (ignore wobble base pairing).
- 3'-UAC-5'
 - 3'-AUG-5'
 - 5'-ATG-3'
 - 5'-AUC-3'
 - 5'-AUG-3'
45. What is on the 3' end of all functional, mature tRNAs?
- The cloverleaf loop
 - The anticodon
 - The sequence CCA
 - The codon
46. Most aminoacyl-tRNA synthetases possess an activity that is shared with DNA polymerases. This activity is a _____ function.
- Proofreading
 - Hydrogenase
 - Proteolytic
 - Helicase
 - Endonucleolytic
47. Which of the following is the CORRECT order for the three distinct phases of protein synthesis?
- Initiation, termination, elongation
 - Termination, initiation, elongation
 - Initiation, elongation, termination
 - Elongation, initiation, termination
 - Elongation, termination, initiation
48. Which amino acid is the initiating amino acid for essentially all proteins?
- Cysteine
 - Threonine
 - Tryptophan
 - Methionine
 - Glutamic acid
49. The initiator tRNA is placed within the active 80S complex at which of the three-canonical ribosomal "sites" during protein synthesis?
- E site
 - I site
 - P site
 - A site
 - Releasing factor binding site
50. Name the enzyme that forms the peptide bond during protein synthesis and define its chemical composition.
- Pepsynthase, protein
 - Peptidyl transferase, RNA
 - Peptidase, glycolipid
 - Peptidyl transferase, protein
 - GTPase, glycopeptide
51. What is the term used for mutations in the middle of an open reading frame that create a stop codon?
- Frameshift mutation
 - Missense mutation
 - No-nonsense mutation
 - Point mutation
 - Nonsense mutation
52. What is the directionality of polypeptide synthesis?
- C-terminal to N-terminal direction
 - N-terminal to 3' direction
 - N-terminal to C-terminal direction
 - 3' to 5' direction
 - 5' to 3' direction

53. Which of the following *cis*-acting elements typically resides adjacent to or overlaps with many prokaryotic promoters?
- Regulatory gene
 - Structural gene(s)
 - Repressor
 - Operator
 - Terminator
54. What is the term applied to a segment of a bacterial chromosome where genes for the enzymes of a particular metabolic pathway are clustered and subject to coordinate control?
- Operon
 - Operator
 - Promoter
 - Terminal controller
 - Origin
55. What is the term applied to the complete collection of proteins present in a particular cell type?
- Genome
 - Peptide collection
 - Transcriptome
 - Translatome
 - Proteome
56. How does nucleosome formation on genomic DNA affect the initiation and/or elongation phases of transcription?
- Nucleosomes inhibit access of enzymes involved in all phases of transcription.
 - Nucleosomes recruit histone and DNA-modifying enzymes, and the actions of these recruited enzymes affect the access of transcription proteins to DNA.
 - Nucleosomes induce DNA degradation where the DNA contacts the histones.
 - Nucleosomes have no significant effect on transcription.
57. Which types of molecules interact with eukaryotic mRNA gene core promoter sites to facilitate the association of RNA polymerase II?
- Termination factors
 - Sequence-specific transcription factors (transactivators)
 - Elongation factors
 - GTPases
 - General, or basal transcription factors (ie, the GTFs)
58. Most eukaryotic transcription factors contain at least two domains, each of which mediate different aspects of transcription factor function. Which of the following are these domains?
- RNA-binding domain and repression domain
 - Activation domain and repression domain
 - DNA-binding domain and activation domain
 - DNA-binding domain and ligand-binding domain
 - RNA-binding domain and the activation domain
59. Transcription factors bound at enhancers stimulate the initiation of transcription at the *cis*-linked core promoter through the action of intermediaries are termed as which of the following?
- Coactivators
 - Cotranscription proteins
 - Corepressors
 - Receptors
 - Coordinators
60. What reactions among transcription proteins greatly expand the diversity of regulatory factors that can be generated from a small number of polypeptides?
- Recombination
 - Homodimerization
 - Heterozygosity
 - Heterodimerization
 - Trimerization
61. The gene region containing the TATA box and extending to the transcription start site (TSS) is often termed which of the following?
- Polymerase home
 - Initiator
 - Start selector
 - Core promoter
 - Operator
62. Which of the following possible mechanisms for how enhancers can stimulate transcription from great distances are currently thought to be CORRECT?
- Enhancers can reversibly excise the intervening DNA between enhancers and promoters.
 - RNA polymerase II binds avidly to enhancer sequences.
 - Enhancers unwind DNA.
 - Enhancers can search through DNA and bind directly to the associated core promoter.
 - Enhancers and core promoters are brought into close proximity through DNA loop formation mediated by DNA-binding proteins.
63. Which of the following histone amino acids are typically acetylated?
- Lysine
 - Arginine
 - Asparagine
 - Histidine
 - Leucine
64. Place the following steps in order; what are the steps that occur sequentially during a transcription activation event following the binding of a transcriptional activator to its cognate activator-binding site on genomic DNA.
- The chromatin remodeling complex binds to the core histones at the target region.
 - The combined actions of the various molecular complexes increase promoter accessibility to the transcriptional machinery.
 - The activator recruits a coactivator to a region of chromatin targeted for transcription.
 - Transcriptional machinery assembles at the site where transcription will be initiated.
 - The coactivator acetylates the core histones of nearby nucleosomes.
- 1 - 2 - 3 - 4 - 5
 - 3 - 1 - 5 - 2 - 4
 - 3 - 5 - 1 - 2 - 4
 - 5 - 3 - 1 - 2 - 4
 - 3 - 5 - 1 - 4 - 2

65. What strategy in transcription factor research allows for the simultaneous identification of all of the genomic sites bound by a given transcription factor under a given set of physiologic conditions?
- A. Systematic deletion mapping
 - B. DNase I sensitivity
 - C. Chromatin immunoprecipitation-sequencing (ChIP-seq)
 - D. FISH
 - E. Fluorescence lifetime imaging microscopy
66. Which sequences extend between the 5'-methylguanosine cap present on eukaryotic mRNAs to the AUG initiation codon?
- A. Stop codon
 - B. Last exon
 - C. Last intron
 - D. 3' UTR
 - E. 5' UTR
67. Which of the following features of eukaryotic mRNA contribute importantly to message half-life?
- A. 5'-UTR sequences
 - B. The promoter
 - C. The operator
 - D. 3' UTR and poly(A) tail
 - E. The first intron



Exam QUESTIONS

Section VIII – Biochemistry of Extracellular & Intracellular Communication

1. Regarding membrane lipids, select the one FALSE answer.
 - A. The major phospholipid by mass in human membranes is generally phosphatidylcholine.
 - B. Glycolipids are located on the inner and outer leaflets of the plasma membrane.
 - C. Phosphatidic acid is a precursor of phosphatidylserine, but NOT of sphingomyelin.
 - D. Phosphatidylcholine and phosphatidylethanolamine are located primarily on the outer leaflet of the plasma membrane.
 - E. The flip-flop of phospholipids in membranes is very slow.
2. Regarding membrane proteins, select the one FALSE answer.
 - A. Because of steric considerations, α -helices cannot exist in membranes.
 - B. A hydropathy plot helps one to estimate whether a segment of a protein is predominantly hydrophobic or hydrophilic.
 - C. Certain proteins are anchored to the outer leaflet of plasma membranes via glycosylphosphatidylinositol (GPI) structures.
 - D. Adenyl cyclase is a marker enzyme for the plasma membrane.
 - E. Myelin has a very high content of lipid compared with protein.
3. Regarding membrane transport, select the one FALSE statement.
 - A. Potassium has a lower charge density than sodium and tends to move more quickly through membranes than does sodium.
 - B. The flow of ions through ion channels is an example of passive transport.
 - C. Facilitated diffusion requires a protein transporter.
 - D. Inhibition of the $\text{Na}^+\text{-K}^+\text{-ATPase}$ will inhibit sodium-dependent uptake of glucose in intestinal cells.
 - E. Insulin, by recruiting glucose transporters to the plasma membrane, increases uptake of glucose in fat cells but not in muscle.
4. Regarding the $\text{Na}^+\text{-K}^+\text{-ATPase}$, select the one FALSE statement.
 - A. Its action maintains the high intracellular concentration of sodium compared with potassium.
 - B. It can use as much as 30% of the total ATP expenditure of a cell.
 - C. It is inhibited by digitalis, a drug that is useful in certain cardiac conditions.
 - D. It is located in the plasma membrane of cells.
 - E. Phosphorylation is involved in its mechanism of action, leading to its classification as a P-type ATP-driven active transporter.
5. What molecules enable cells to respond to a specific extracellular signaling molecule?
 - A. Specific receptor carbohydrates localized to the inner plasma membrane surface
 - B. Plasma lipid bilayer
 - C. Ion channels
 - D. Receptors that specifically recognize and bind that particular messenger molecule
 - E. Intact nuclear membranes
6. Indicate the term generally applied to the extracellular messenger molecules that bind to transmembrane receptor proteins.
 - A. Competitive inhibitor
 - B. Ligand
 - C. Scatchard curve
 - D. Substrate
 - E. Key
7. In autocrine signaling:
 - A. Messenger molecules reach their target cells via passage through bloodstream.
 - B. Messenger molecules travel only short distances through the extracellular space to cells that are in close proximity to the cell that is generating the message.
 - C. The cell producing the messenger expresses receptors on its surface that can respond to that messenger.
 - D. The messenger molecules are usually rapidly degraded and hence can only work over short distances.
8. Regardless of how a signal is initiated, the ligand-binding event is propagated via second messengers or protein recruitment. What is the ultimate, or final biochemical outcome of these binding events?
 - A. A protein in the middle of an intracellular signaling pathway is activated.
 - B. A protein at the bottom of an intracellular signaling pathway is activated.
 - C. A protein at the top of an extracellular signaling pathway is activated.
 - D. A protein at the top of an intracellular signaling pathway is deactivated.
 - E. A protein at the top of an intracellular signaling pathway is activated.
9. What features of the nuclear receptor superfamily suggest that these proteins have evolved from a common ancestor?
 - A. They all bind the same ligand with high affinity.
 - B. They all function within the nucleus.
 - C. They are all subject to regulatory phosphorylation.
 - D. They all contain regions of high amino acid sequence similarity/identity.
 - E. They all bind DNA.
10. What effect does degradation of receptor-ligand complexes after internalization have upon the ability of a cell to respond if immediately reexposed to the same hormone?
 - A. The cellular response is attenuated due to a decrease in cellular receptor number.
 - B. Cellular response is enhanced due to reduced receptor-ligand competition.
 - C. The cellular response is unchanged to subsequent stimuli.
 - D. Cell hormone response is now bimodal; enhanced for a short time and thereafter inactivated.

11. Typically, what is the first reaction after most receptor protein-tyrosine kinases (RTKs) bind their ligand?
 - A. RECEPTOR trimerization
 - B. RECEPTOR degradation
 - C. RECEPTOR denaturation
 - D. RECEPTOR dissociation
 - E. RECEPTOR dimerization
12. Where is the kinase catalytic domain of the receptor protein-tyrosine kinases found?
 - A. ON THE EXTRACELLULAR SURFACE OF THE RECEPTOR, immediately adjacent to the ligand-binding domain.
 - B. ON THE CYTOPLASMIC DOMAIN OF THE RECEPTOR.
 - C. ON AN INDEPENDENT PROTEIN THAT RAPIDLY BINDS THE RECEPTOR upon ligand binding.
 - D. WITHIN THE TRANSMEMBRANE SPANNING PORTION OF THE RECEPTOR.
13. The subunits of the heterotrimeric G-proteins are called the __, __, and __ subunits.
 - A. α , β , and χ
 - B. α , β , and δ
 - C. α , γ , and δ
 - D. α , β , and γ
 - E. γ , δ , and η
14. Of the receptors listed below, which can directly conduct a flow of ions across the plasma membrane when bound to their cognate ligand?
 - A. RECEPTOR TYROSINE KINASES (RTKs)
 - B. G-protein-coupled receptors (GPCRs)
 - C. G-protein gamma a subunit.
 - D. Steroid hormone receptors
 - E. Ligand-gated channels
15. Which of the following is NOT a natural ligand that binds to G-protein-coupled receptors?
 - A. HORMONES
 - B. Steroid hormones
 - C. Chemoattractants
 - D. Opium derivatives
 - E. Neurotransmitters
16. Place the events of signaling listed below in the correct order
 1. G-protein binds to activated receptor forming a receptor-G-protein complex.
 2. Release of GDP by the G-protein.
 3. Change in conformation of the cytoplasmic loops of the receptor.
 4. Binding of GTP by the G-protein.
 5. INCREASE IN THE AFFINITY OF THE RECEPTOR FOR A G-PROTEIN ON THE CYTOPLASMIC SURFACE OF THE MEMBRANE.
 6. Binding of a hormone or neurotransmitter to a G-protein-coupled receptor.
 7. Conformational shift in the α subunit of the G-protein.
 - A. 6 - 3 - 5 - 1 - 2 - 4 - 7
 - B. 6 - 5 - 4 - 1 - 7 - 2 - 3
 - C. 6 - 3 - 5 - 1 - 7 - 2 - 4
 - D. 6 - 7 - 3 - 5 - 1 - 2 - 4
 - E. 6 - 3 - 5 - 4 - 7 - 2 - 1
17. Which heterotrimeric G-proteins couple receptors to adenylyl cyclase via the activation of GTP-bound G_s subunits?
 - A. G_i family
 - B. G_q family
 - C. $G_{12/13}$ family
 - D. $G_{12/13}$ family
 - E. G_s family
18. What must happen in order to prevent overstimulation by a hormone?
 - A. HORMONES MUST BE DEGRADED.
 - B. G-PROTEINS MUST BE RECYCLED AND THEN DEGRADED.
 - C. RECEPTORS MUST BE BLOCKED FROM CONTINUING TO ACTIVATE G-PROTEINS.
 - D. RECEPTORS MUST DIMERIZE.
19. Which of the following hormones termed the "flight-or-fight" hormone is secreted by the adrenal medulla?
 - A. Epinephrine
 - B. Oxytocin
 - C. Insulin
 - D. Glucagon
 - E. Somatostatin
20. Which hormone is secreted by a cells in the pancreas in response to low blood glucose levels?
 - A. Insulin
 - B. Glucagon
 - C. Estradiol
 - D. Epinephrine
 - E. Somatostatin
21. In liver cells, the expression of genes encoding gluconeogenic enzymes such as phosphoenolpyruvate carboxykinase is induced in response to which of the following molecules?
 - A. cGMP
 - B. Insulin
 - C. ATP
 - D. cAMP
 - E. Cholesterol
22. What happens to protein kinase A (PKA) following the binding of cAMP?
 - A. The regulatory subunits of PKA dissociate, thereby activating the catalytic subunits.
 - B. PKA catalytic subunits then bind to two regulatory subunits, thereby activating the catalytic subunits.
 - C. The inhibitory regulatory subunits dissociate from the catalytic subunits, completely inactivating the enzyme.
 - D. The stimulatory regulatory subunits dissociate from the catalytic subunits, inhibiting the enzyme.
 - E. Phosphodiesterase binds to the catalytic subunits, which results in enzyme inactivation.

Exam Questions

Section IX – Special Topics (A)

- Which of the following will be elevated in the bloodstream about 1 to 2 hours after eating a high-fat meal?
 - Chylomicrons
 - High density lipoprotein
 - Ketone bodies
 - Nonesterified fatty acids
 - Very low-density lipoprotein
- Which of the following will be elevated in the bloodstream about 4 to 5 hours after eating a high-fat meal?
 - Chylomicrons
 - High-density lipoprotein
 - Ketone bodies
 - Nonesterified fatty acids
 - Very low-density lipoprotein
- Which of the following is the best definition of glycemic index?
 - The increase in the blood concentration of glucagon after consuming the food compared with that after an equivalent amount of white bread.
 - The increase in the blood concentration of glucose after consuming the food.
 - The increase in the blood concentration of glucose after consuming the food compared with that after an equivalent amount of white bread.
 - The increase in the blood concentration of insulin after consuming the food.
 - The increase in the blood concentration of insulin after consuming the food compared with that after an equivalent amount of white bread.
- Which of the following will have the lowest glycemic index?
 - A baked apple
 - A baked potato
 - An uncooked apple
 - An uncooked potato
 - Apple juice
- Which of the following will have the highest glycemic index?
 - A baked apple
 - A baked potato
 - An uncooked apple
 - An uncooked potato
 - Apple juice
- Which one of the following statements concerning chylomicrons is CORRECT?
 - Chylomicrons are made inside intestinal cells and secreted into lymph, where they acquire apolipoproteins B and C.
 - The core of chylomicrons contains triacylglycerol and phospholipids.
 - The enzyme hormone-sensitive lipase acts on chylomicrons to release fatty acids from triacylglycerol when they are bound to the surface of endothelial cells in blood capillaries.
 - Chylomicron remnants differ from chylomicrons in that they are smaller and contain a lower proportion of triacylglycerol.
 - Chylomicrons are taken up by the liver.
- Plant sterols and stanols inhibit the absorption of cholesterol from the gastrointestinal tract. Which of the following best describes how they act?
 - They are incorporated into chylomicrons in place of cholesterol.
 - They compete with cholesterol for esterification in the intestinal lumen, so that less cholesterol is esterified.
 - They compete with cholesterol for esterification in the mucosal cell, and unesterified cholesterol is actively transported out of the cell into the intestinal lumen.
 - They compete with cholesterol for esterification in the mucosal cell, and unesterified cholesterol is not incorporated into chylomicrons.
 - They displace cholesterol from lipid micelles, so that it is not available for absorption.
- Which one of following statements about energy metabolism is CORRECT?
 - Adipose tissue does not contribute to basal metabolic rate (BMR).
 - Physical activity level (PAL) is the sum of physical activity ratios for different activities throughout the day, multiplied by the time spent in each activity, expressed as a multiple of BMR.
 - Physical activity ratio (PAR) is the energy cost of physical activity throughout the day.
 - Resting metabolic rate (RMR) is the energy expenditure of the body when asleep.
 - The energy cost of physical activity can be determined by measuring respiratory quotient (RQ) production during the activity.
- A patient with metastatic colorectal cancer has lost 6 kg of body weight over the last month. Which of the following is the best explanation for her weight loss?
 - Because of the tumour she is oedematous.
 - Chemotherapy has caused nausea and loss of appetite.
 - Her basal metabolic rate has fallen as a result of protein catabolism caused by tumour necrosis factor and other cytokines.
 - Her basal metabolic rate (BMR) has increased as a result of anaerobic glycolysis in the tumour and the energy cost of gluconeogenesis from the resultant lactate in her liver.
 - The tumour has a very high energy requirement for cell proliferation.

10. A 5-year-old child arriving at a refugee center in East Africa is stunted in growth (only 89% of expected height for age) but not oedematous. Would you consider him to be:
- Suffering from kwashiorkor
 - Suffering from marasmic kwashiorkor
 - Suffering from marasmus
 - Suffering from undernutrition
 - Underfed but not considered to be clinically malnourished
11. A 5-year-old child arriving at a refugee center in East Africa is stunted in growth (only 55% of expected height for age) but not oedematous. Would you consider him to be:
- Suffering from kwashiorkor
 - Suffering from marasmic kwashiorkor
 - Suffering from marasmus
 - Suffering from undernutrition
 - Underfed but not considered to be clinically malnourished
12. Which of the following is the definition of nitrogen balance?
- Protein intake as a percentage of total energy intake
 - The difference between protein intake and excretion of nitrogenous compounds
 - The ratio of excretion of nitrogenous compounds/protein intake
 - The ratio of protein intake/excretion of nitrogenous compounds
 - The sum of protein intake and excretion of nitrogenous compounds
13. Which one of following statements about nitrogen balance is CORRECT?
- If the intake of protein is greater than requirements, there will always be positive nitrogen balance.
 - In nitrogen equilibrium, the excretion of nitrogenous metabolites is greater than the dietary intake of nitrogenous compounds.
 - In positive nitrogen balance, the excretion of nitrogenous metabolites is less than the dietary intake of nitrogenous compounds.
 - Nitrogen balance is the ratio of intake of nitrogenous compounds/output of nitrogenous metabolites from the body.
 - Positive nitrogen balance means that there is a net loss of protein from the body.
14. In a series of experiments to determine amino acid requirements, healthy young adult volunteers were fed mixtures of amino acids as their sole protein source. Which of the following mixtures would lead to negative nitrogen balance (assuming that all other amino acids are provided in adequate amounts)?
- One lacking alanine, glycine, and tyrosine
 - One lacking arginine, glycine, and cysteine
 - One lacking asparagine, glutamine, and cysteine
 - One lacking lysine, glycine, and tyrosine
 - One lacking proline, alanine, and glutamate
15. Which of the following vitamins provides the cofactor for reduction reactions in fatty acid synthesis?
- Folate
 - Niacin
 - Riboflavin
 - Thiamin
 - Vitamin B₆
16. Deficiency of which one of these vitamins is a major cause of blindness worldwide?
- Vitamin A
 - Vitamin B₁₂
 - Vitamin B₆
 - Vitamin D
 - Vitamin K
17. Deficiency of which one of these vitamins may lead to megaloblastic anaemia?
- Vitamin B₆
 - Vitamin B₁₂
 - Vitamin D
 - Vitamin E
 - Vitamin K
18. Which one of the following criteria of vitamin adequacy can be defined as "There are no signs of deficiency under normal conditions, but any trauma or stress reveals the precarious state of the body reserves and may precipitate clinical signs"?
- Abnormal response to a metabolic load
 - Clinical deficiency disease
 - Covert deficiency
 - Incomplete saturation of body reserves
 - Subclinical deficiency
19. Which one of the following criteria of vitamin adequacy can be defined as metabolic abnormalities under normal conditions?
- Abnormal response to a metabolic load
 - Clinical deficiency disease
 - Covert deficiency
 - Incomplete saturation of body reserves
 - Subclinical deficiency
20. Which of the following is the best definition of the reference nutrient intake (RNI) or recommended daily amount (RDA), of a vitamin or mineral?
- One standard deviation above the average requirement of the population group under consideration
 - One standard deviation below the average requirement of the population group under consideration
 - The average requirement of the population group under consideration
 - Two standard deviations above the average requirement of the population group under consideration
 - Two standard deviations below the average requirement of the population group under consideration
21. What percentage of the population will have met their requirement for a vitamin or mineral if their intake is equal to the RNI or RDA?
- 2.5%
 - 5%
 - 50%
 - 95%
 - 97.5%

22. What percentage of the population will have met their requirement for a vitamin or mineral if their intake is equal to the lower reference nutrient intake (LRNI)?
- 2.5%
 - 5%
 - 50%
 - 95%
 - 97.5%
23. What percentage of the population will have met their requirement for a vitamin or mineral if their intake is equal to the average requirement?
- 2.5%
 - 5%
 - 50%
 - 95%
 - 97.5%
24. For a person whose intake of a vitamin or mineral is equal to the average requirement, what is the probability that this level of intake is adequate to meet his/her individual requirement?
- 2.5%
 - 5%
 - 50%
 - 95%
 - 97.5%
25. For a person whose intake of a vitamin or mineral is equal to the LRNI, what is the probability that this level of intake is adequate to meet his/her individual requirement?
- 2.5%
 - 5%
 - 50%
 - 95%
 - 97.5%
26. For a person whose intake of a vitamin or mineral is equal to the RNI, what is the probability that this level of intake is adequate to meet his/her individual requirement?
- 2.5%
 - 5%
 - 50%
 - 95%
 - 97.5%
27. Which one of the following is NOT a source of oxygen radicals?
- Action of superoxide dismutase
 - Activation of macrophages
 - Nonenzymic reactions of transition metal ions
 - Reaction of β -carotene with oxygen
 - Ultraviolet radiation
28. Which one of the following provides protection against oxygen radical damage to tissues?
- Action of superoxide dismutase
 - Activation of macrophages
 - Nonenzymic reactions of transition metal ions
 - Reaction of β -carotene with oxygen
 - Ultraviolet radiation
29. Which one of the following is NOT the result of oxygen radical action?
- Activation of macrophages
 - Modification of bases in DNA
 - Oxidation of amino acids in apoproteins of LDL
 - Peroxidation of unsaturated fatty acids in membranes
 - Strand breaks in DNA
30. Which of the following types of oxygen radical damage may lead to the development of autoimmune thyroid disease?
- Chemical modification of DNA bases in somatic cells
 - Chemical modification of DNA in germ-line cells
 - Oxidation of amino acids in cell membrane proteins
 - Oxidation of amino acids in mitochondrial proteins
 - Oxidation of unsaturated fatty acids in plasma lipoproteins
31. Which of the following types of oxygen radical damage may lead to the development of atherosclerosis and coronary heart disease?
- Chemical modification of DNA bases in somatic cells
 - Chemical modification of DNA in germ-line cells
 - Oxidation of amino acids in cell membrane proteins
 - Oxidation of amino acids in mitochondrial proteins
 - Oxidation of unsaturated fatty acids in plasma lipoproteins
32. Which of the following types of oxygen radical damage may lead to the development of cancer?
- Chemical modification of DNA bases in somatic cells
 - Chemical modification of DNA in germ-line cells
 - Oxidation of amino acids in cell membrane proteins
 - Oxidation of amino acids in mitochondrial proteins
 - Oxidation of unsaturated fatty acids in plasma lipoproteins
33. Which of the following types of oxygen radical damage may lead to the development of hereditary mutations?
- Chemical modification of DNA bases in somatic cells
 - Chemical modification of DNA in germ-line cells
 - Oxidation of amino acids in cell membrane proteins
 - Oxidation of amino acids in mitochondrial proteins
 - Oxidation of unsaturated fatty acids in plasma lipoproteins
34. Which one of the following best explains the antioxidant action of vitamin E?
- It forms a stable radical that can be reduced back to active vitamin E by reaction with vitamin C.
 - It is a radical, so that when it reacts with another radical, a nonradical product is formed.
 - It is converted to a stable radical by reaction with vitamin C.
 - It is lipid soluble and can react with free radicals in the blood plasma resulting from nitric oxide (NO) formation by vascular endothelium.
 - Oxidized vitamin E can be reduced back to active vitamin E by reaction with glutathione and glutathione peroxidase.
35. Which of the following best describes the glycome?
- The DNA coding for glycosyltransferases
 - The full complement of all carbohydrates in the body
 - The full complement of free sugars in cells and tissues
 - The full complement of glycoproteins and glycolipids in the body
 - The full complement of glycosyltransferases in the body
36. Which of the following methods CANNOT be used to determine the structures of glycoproteins?
43. Which one of the following statements is INCORRECT?
- Calnexin ensures the correct folding of glycoproteins in the

36. Which of the following methods CANNOT be used to determine the structures of glycoproteins?
- Carbohydrate microarrays
 - Degradation using endo- and exoglycosidases
 - Genome analysis
 - Mass spectrometry
 - Sepharose-lectin chromatography
37. Which of the following is NOT a function of glycoproteins?
- Anchoring proteins at the cell surface
 - Protecting plasma proteins against clearance by the liver
 - Providing a transport system for folate into cells
 - Providing a transport system for uptake of low-density lipoprotein into the liver
 - Providing cell surface recognition signals
38. Which of the following is NOT a constituent of glycoproteins?
- Fucose
 - Galactose
 - Glucose
 - Mannose
 - Sucrose
39. Which of the following is used as a sugar donor in the synthesis of the common pentasaccharide of *N*-linked glycoproteins?
- CMP-*N*-acetylneuraminic acid
 - Dolichol pyrophosphate *N*-acetylglucosamine
 - Dolichol pyrophosphate-mannose
 - GDP-fucose
 - UDP-*N*-acetylglucosamine
40. Which of the following is NOT used as a sugar donor in the synthesis of *N*-linked glycoproteins in the endoplasmic reticulum?
- Dolichol pyrophosphate fructose
 - Dolichol pyrophosphate galactose
 - Dolichol pyrophosphate mannose
 - Dolichol pyrophosphate *N*-acetylglucosamine
 - Dolichol pyrophosphate *N*-acetylneuraminic acid
41. Which of the following best describes the attachment of the common pentapeptide to the apoprotein in synthesis of an *N*-linked glycoprotein?
- Direct glycation of the amino terminal amino acid of the peptide
 - Displacement of the amino terminal region of the peptide in a transamidation reaction
 - Displacement of the amino terminal region of the peptide in a transamination reaction
 - Displacement of the carboxy terminal region of the peptide in a transamidation reaction
 - Displacement of the carboxy terminal region of the peptide in a transamination reaction
42. Which of the following is NOT a glycoprotein?
- Collagen
 - Immunoglobulin G
 - Serum albumin
 - Thyroid-stimulating hormone
 - Transferrin
43. Which one of the following statements is INCORRECT?
- Calnexin ensures the correct folding of glycoproteins in the endoplasmic reticulum.
 - Dolichol-pyrophosphate oligosaccharide donates all of the sugars found in *N*-linked glycoproteins.
 - Mucins contain predominantly *O*-linked glycans.
 - N*-Acetylneuraminic acid is commonly found at the termini of *N*-linked sugar chains of glycoproteins.
 - O*-linked sugar chains of glycoproteins are built up by the stepwise addition of sugars from sugar nucleotides.
44. Which of the following is NOT an activity of cytochrome P450?
- Activation of vitamin D
 - Hydroxylation of steroid hormone precursors
 - Hydroxylation of xenobiotics
 - Hydroxylation of retinoic acid
 - Methylation of xenobiotics
45. Which of the following best describes the reaction of a cytochrome P450?
- $\text{RH} + \text{O}_2 + \text{NADP}^+ \rightarrow \text{R-OH} + \text{H}_2\text{O} + \text{NADPH}$
 - $\text{RH} + \text{O}_2 + \text{NAD}^+ \rightarrow \text{R-OH} + \text{H}_2\text{O} + \text{NADH}$
 - $\text{RH} + \text{O}_2 + \text{NADPH} \rightarrow \text{R-OH} + \text{H}_2\text{O} + \text{NADP}^+$
 - $\text{RH} + \text{O}_2 + \text{NADPH} \rightarrow \text{R-OH} + \text{H}_2\text{O}_2 + \text{NADP}^+$
 - $\text{RH} + \text{O}_2 + \text{NADH} \rightarrow \text{R-OH} + \text{H}_2\text{O} + \text{NAD}^+$
46. Which of the following is the preferred lipid component of the cytochrome P450 system?
- Dolichol phosphate
 - Phosphatidylcholine
 - Phosphatidylethanolamine
 - Phosphatidylinositol
 - Phosphatidylserine
47. Which of the following best describes the drug interactions between phenobarbital and warfarin?
- Phenobarbital induces CYP2C9, and this results in decreased catabolism of warfarin.
 - Phenobarbital induces CYP2C9, and this results in increased catabolism of warfarin.
 - Phenobarbital represses CYP2C9, and this results in increased catabolism of warfarin.
 - Warfarin induces CYP2C9, and this results in decreased catabolism of phenobarbital.
 - Warfarin induces CYP2C9, and this results in increased catabolism of phenobarbital.

48. Which of the following best describes the effects of polymorphisms of CYP2A6?
- A. People with the active allele are less likely to become tobacco-dependent smokers because this cytochrome inactivates nicotine to cotinine.
 - B. People with the inactive (null) allele are less likely to become tobacco-dependent smokers because this cytochrome inactivates nicotine to cotinine.
 - C. People with the inactive (null) allele are less likely to become tobacco-dependent smokers because this cytochrome activates nicotine to cotinine.
 - D. People with the inactive (null) allele are more likely to become tobacco-dependent smokers because this cytochrome inactivates nicotine to cotinine.
 - E. People with the inactive (null) allele are more likely to become tobacco-dependent smokers because this cytochrome activates nicotine to cotinine.
49. Which of the following is NOT a function of glutathione?
- A. Coenzyme for the reduction of hydrogen peroxide
 - B. Conjugation of bilirubin
 - C. Conjugation of some products of phase I metabolism of xenobiotics
 - D. Transport of amino acids across cell membranes
 - E. Transport of bilirubin in the bloodstream
50. Which of the following best describes the reference range for a laboratory test?
- A. A range $\pm 1 \times$ standard deviation around the mean value
 - B. A range $\pm 1.5 \times$ standard deviation around the mean value
 - C. A range $\pm 2 \times$ standard deviation around the mean value
 - D. A range $\pm 2.5 \times$ standard deviation around the mean value
 - E. A range $\pm 3 \times$ standard deviation around the mean value
51. Which of the following statements about laboratory tests is INCORRECT?
- A. The predictive value of a test is the extent to which it will correctly predict whether or not a person has the disease.
 - B. The sensitivity and specificity of a test are inversely related.
 - C. The sensitivity of a test is a measure of how many people with the disease will give a positive result.
 - D. The specificity of a test is a measure of how many people with the disease will give a positive result.
 - E. The specificity of a test is a measure of how many people without the disease will give a negative result.
52. Which of the following is CORRECT when an enzyme is used to measure an analyte in a blood sample?
- A. The concentration of substrate must be about 20-times the K_m of the enzyme.
 - B. The concentration of substrate must be equal to the K_m of the enzyme.
 - C. The concentration of substrate must be equal to or lower than the K_m of the enzyme.
 - D. The concentration of the substrate in the assay is not important.
 - E. The concentration of substrate must be about 1/20th of the K_m of the enzyme.
53. Which of the following is CORRECT when an enzyme is being measured in a blood sample?
- A. The concentration of substrate must be about 20 times the K_m of the enzyme.
 - B. The concentration of substrate must be equal to the K_m of the enzyme.
 - C. The concentration of substrate must be equal to or lower than the K_m of the enzyme.
 - D. The concentration of the substrate in the assay is not important.
 - E. The concentration of substrate must be about 1/20th of the K_m of the enzyme.
54. Which of the following best explains the use of enzyme activation assays to assess vitamin nutritional status?
- A. Adding the vitamin-derived cofactor to the incubation converts previously inactive apoenzyme into active holoenzyme.
 - B. Adding the vitamin-derived cofactor to the incubation converts previously inactive holoenzyme into active apoenzyme.
 - C. Adding the vitamin-derived cofactor to the incubation converts previously active holoenzyme into inactive apoenzyme.
 - D. Adding the vitamin-derived cofactor to the incubation converts previously active apoenzyme into inactive holoenzyme.
 - E. Adding the vitamin-derived cofactor to the incubation leads to a reduction in enzyme activity.
55. Which of the following would be used to prepare serum from a blood sample?
- A. A plain tube
 - B. A tube containing citrate
 - C. A tube containing EDTA
 - D. A tube containing oxalate
 - E. An evacuated tube to exclude oxygen
56. Which of the following would be used to take a blood sample for blood gas analysis?
- A. A plain tube
 - B. A tube containing citrate
 - C. A tube containing EDTA
 - D. A tube containing oxalate
 - E. An evacuated tube to exclude oxygen
57. Which of the following best explains the difference between creatinine clearance and inulin clearance as tests of renal function?
- A. Creatinine clearance is higher than inulin clearance because creatinine is actively secreted in the distal renal tubules.
 - B. Creatinine clearance is higher than inulin clearance because inulin is actively secreted in the proximal renal tubules.
 - C. Creatinine clearance is higher than inulin clearance because inulin is actively secreted in the distal renal tubules.
 - D. Creatinine clearance is lower than inulin clearance because creatinine is actively secreted in the distal renal tubules.
 - E. Creatinine clearance is lower than inulin clearance because inulin is not completely filtered at the glomerulus.

Exam Questions

Section X – Special Topics (B)

- Briefly describe the mode of action of nitroglycerin, a common agent for treating angina.
- Patients being treated for heart failure oftentimes exhibit decreased expression and defective regulation of SERCA2a, the principal Ca^{2+} -ATPase of the sarcoplasmic reticulum. Explain how defects in this protein might contribute to deterioration in cardiac function.
- Select the one of the following that is NOT CORRECT:
 - The troponin system regulates contraction of smooth muscle.
 - Muscle contraction takes place via a sliding filament mechanism.
 - Myosin light-chain kinase phosphorylates the regulatory light chains in the myosin head domain.
 - F-actin is formed via the polymerization of G-actin.
 - Ca^{2+} both activates muscle contraction and stimulates its own removal by activating the Ca^{2+} ATPase.
- A patient anesthetized using a halothane compound exhibits a marked rise in body temperature, a behavior indicative of malignant hyperthermia (HT). Select the one of the following statements that is NOT CORRECT:
 - MH can arise from mutations that alter the amino acid sequence of the Na^+ - K^+ -ATPase.
 - MH can arise from mutations that alter the amino acid sequence of the ryanodine-sensitive Ca^{2+} release channel.
 - The muscle rigidity that occurs during MH is triggered by the presence of high concentrations of Ca^{2+} in the cytoplasm.
 - MH can arise from mutations that alter the amino acid sequence of the voltage-gated, slow K-type Ca^{2+} channel.
 - MH can be treated by intravenous administration of dantrolene, which inhibits release of Ca^{2+} from the sarcoplasmic reticulum into the cytosol.
- Select the one of the following statements that is NOT CORRECT:
 - Fast-twitch fibers rely heavily on creatine phosphate to regenerate ATP.
 - Slow-twitch fibers appear red because they contain hemoglobin.
 - Fast-twitch fibers contain relatively few mitochondria.
 - Marathoners try to increase the quantity of glycogen in their muscles by eating carbohydrate-rich meals before an event (carbo loading).
 - Skeletal muscle serves as the major reserve of protein in the body.
- Select the one of the following that is NOT a feature of the contractile cycle in striated muscle:
 - Binding of Ca^{2+} to troponin C uncovers the myosin-binding sites on actin.
 - The power stroke is initiated by the release of P_i from the actin-myosin-ADP- P_i complex.
 - Release of ADP from the actin-myosin-ADP complex is accompanied by a large change in the conformation of myosin's head domain (relative to its tail domain).
 - The binding of ATP by myosin increases its affinity for actin.
 - Rigor mortis results from the inability of actin to release from the actin-myosin complex when cells are deficient in ATP.
- Select the one of the following that does NOT serve as a major energy reserve for replenishing ATP in muscle tissue:
 - Glycogen
 - Creatine phosphate
 - ADP (in conjunction with adenylate kinase)
 - Fatty acids
 - Epinephrine
- Select the one of the following statements that is NOT CORRECT:
 - The drugs colchicine and vinblastine inhibit microtubule assembly.
 - Mutations affecting keratin can lead to blistering.
 - Mutations in the gene encoding lamin A and lamin C cause progeria (accelerated aging).
 - α - and β -tubulin are the major components of stress fibers.
 - Molecular motors such as dynein, kinesin, and dynamin power ciliary movement, vesicle transport, and endocytosis.
- Select the one of the following statements that is NOT CORRECT:
 - The major function of the Ca^{2+} channels in cardiomyocytes is to admit extracellular calcium ions into the cell in order to trigger Ca^{2+} -induced Ca^{2+} release from the SR.
 - Digitalis increases the forcefulness of cardiac contractions by raising the level of intracellular Na^+ .
 - Certain types of muscular dystrophies are caused by mutations in enzymes called glycosyltransferases.
 - Dantrolene relaxes skeletal muscle by inhibiting the release of Ca^{2+} from the SR.
 - In the SR, Ca^{2+} is bound to a specific Ca^{2+} -binding protein called calmodulin.
- Describe the role of haptoglobin in the protection of the kidneys from the potentially damaging effects of extracorporeal hemoglobin.
- Briefly describe how activation of cytidine deaminase helps generate immunoglobulins with unique antigen-binding sites.
- Select the one of the following statements that is NOT CORRECT:
 - Interleukin 1 stimulates the production of acute-phase proteins.
 - Iron must be reduced to the ferrous (Fe^{2+}) state in order to be recovered via the transferrin cycle.
 - Many complement proteins are zymogens.
 - The type 2 transferrin receptor (TfR2) functions primarily as an iron sensor.
 - Mannose-binding lectin binds carbohydrate groups present on the surface of invading bacteria.

13. Select the one of the following statements that is NOT CORRECT:
- Albumin is synthesized as a proprotein.
 - Albumin is stabilized by multiple intrachain disulfide bonds.
 - Albumin is a glycoprotein.
 - Albumin facilitates the movement of fatty acids through the circulation.
 - Albumin is the major determinant of plasma osmotic pressure.
14. Select the one of the following statements that is NOT CORRECT:
- Wilson disease can be treated using copper chelators such as penicillamine.
 - Wilson disease is characterized by copper toxicosis (abnormally high levels of copper).
 - Wilson disease is caused by mutations in the gene encoding ceruloplasmin.
 - Albumin facilitates the movement of sulfonamide drugs through the circulation.
 - Albumin can be lost from the body if the intestinal mucosa becomes inflamed.
15. You encounter a 50-year-old woman in the clinic who is pale and tired. You suspect that she is suffering from iron deficiency anemia and prescribe a series of laboratory tests. Select the one of the following potential test outcomes that would NOT be consistent with your provisional diagnosis:
- Lower than normal levels of red cell protoporphyrin
 - Increased saturation of transferrin
 - Increased expression of TfR
 - Increased levels of plasma hepcidin
 - Decreased levels of hemoglobin
16. Select the one of the following that is NOT a potential cause of amyloidosis:
- Accumulation of β_2 -macroglobulin
 - Deposition of fragments derived from immunoglobulin light chains
 - Accumulation of degradation products of serum amyloid A
 - Presence of mutationally altered forms of transthyretin
 - Amylase deficiency
17. Select the one of the following statements that is NOT CORRECT:
- All immunoglobulins contain at least two heavy-chain polypeptides and two light-chain polypeptides.
 - Immunoglobulin polypeptide chains are linked together by disulfide bonds.
 - Immunoglobulins are multivalent.
 - Immunoglobulins are glycosylated.
 - Immunoglobulins are primary components of the body's innate immune system.
18. Explain the linkage how a deficiency in glucose-6-phosphate dehydrogenase within erythrocytes can lead to hemolytic anemia.
19. Select the one of the following statements that is NOT CORRECT:
- The high surface area of biconcave red blood cells facilitates gas exchange.
 - Hereditary elliptocytosis can be caused by defects in or a deficiency of spectrin.
 - The diameter of red blood cells exceeds that of many peripheral capillaries.
 - Protein 4.1 helps link the erythrocyte cytoskeleton to proteins in the cell's plasma membrane.
 - In order to pass through narrow capillaries, red blood cells must be squeezed into a compact, spherical shape.
20. Select the one of the following statements that is NOT CORRECT:
- Red blood cells contain high levels of superoxide dismutase.
 - A and B substances are formed by the addition of fucose and N-acetylglucosamine, respectively, to H substance.
 - Platelets generate ATP exclusively via glycolysis.
 - Mature red blood cells are devoid of internal organelles.
 - Erythrocyte membranes contain high levels of the Band 3 anion exchange protein.
21. Select the one of the following statements that is NOT CORRECT:
- Erythropoietin stimulates the formation of red blood cells from hematopoietic stem cells.
 - Multipotent stem cells are able to differentiate into cells of a closely related type.
 - Carbonic anhydrase increases the capacity of red blood cells to transport CO_2 .
 - GLUT1 mediates the active transport of glucose into erythrocytes.
 - Hypoxia stimulates the production of erythropoietin by the kidneys.
22. A patient recently exposed to aniline displays bluish discoloration of their skin and mucous membranes. Select a plausible provisional diagnosis from the list below:
- Methemoglobinemia
 - Hereditary hemochromatosis
 - 5q-syndrome
 - Immune thrombocytopenic purpura
 - Glanzmann thrombasthenia
23. Select the one of the following statements that is NOT CORRECT:
- The accumulation of fluid at a site of infection (edema) facilitates leukocyte migration.
 - Type 1 leukocyte adhesion deficiency is caused a lack of the β_2 subunit an integrin designated LFA-1.
 - The components of the complement cascade circulate through the plasma as inactive zymogens.
 - Leukocytes are recruited to a site of infection by chemotaxis toward the sources of epinephrine.
 - Neutrophils can trap large pathogens in NETs constructed, in part, from strands of chromosomal DNA.
24. Select the one of the following statements that is NOT CORRECT:
- Interleukins are key mediators of leukocyte production.
 - Lymphocytes produce protective antibodies.
 - Monocytes can be found in tissues throughout the body.
 - The hematologic factor histamine is synthesized by the deamination of the amino acid histidine.
 - The term polymorphonuclear refers to leukocytes possessing a segmented nucleus.

25. Select the one of the following statements that is NOT CORRECT:
- Phagocytes destroy ingested bacteria using reactive oxygen species and hydrolytic enzymes.
 - Chronic granulomatous disease is caused by a deficiency in myeloperoxidase activity.
 - NADPH serves as the primary source of electrons for generating ROS during the oxidative burst.
 - Neutrophils aid in the elimination of some parasites by enmeshing them in NETs formed from their chromosomal DNA.
 - Chemokines are stabilized by the formation of intrachain disulfide bonds.
26. Select the one of the following statements that is NOT CORRECT:
- Activated leukocytes secrete lipid mediators called interferons.
 - Neutrophils facilitate production of protective antibodies by presenting fragments of phagocytized microbes on their surface in association with the major histocompatibility complex (MHC).
 - Cytotoxic T cells use perforins to lyse infected cells.
 - Soluble antibodies are released into the plasma primarily by B lymphocytes.
 - Emphysema can arise from the action of elastase and other granule-derived proteases on pulmonary tissue.
27. Select the one FALSE statement:
- The great majority of mitochondrial proteins are encoded by the nuclear genome.
 - Ran proteins, like ARF and Ras proteins, are monomeric GTPases.
 - One cause of Refsum disease is mutations in genes encoding peroxisomal proteins.
 - Peroxisomal proteins are synthesized on cytosolic polyribosomes.
 - Import of proteins into mitochondria involves proteins known as importins.
28. Select the one FALSE statement:
- N-terminal signal peptides directing nascent proteins to the ER membrane contain a hydrophobic sequence.
 - Posttranslational translocation of proteins to the ER does not occur in mammalian species.
 - The SRP contains one RNA species.
 - N-glycosylation is catalyzed by oligosaccharyl transferase.
 - Type I membrane proteins have their N-termini facing the lumen of the ER.
29. Select the one FALSE statement:
- Chaperones often exhibit ATPase activity.
 - Protein disulfide isomerase and peptidyl prolyl isomerase are enzymes involved in helping proteins fold properly.
 - Ubiquitin is a small protein involved in protein degradation by lysosomes.
 - Mitochondria contain chaperones.
 - Retrotranslocation across the ER membrane is involved in helping dispose of misfolded proteins.
30. Select the one FALSE statement:
- Rab is a small GTPase involved in vesicle targeting.
 - COPII vesicles are involved in anterograde transport of cargo from the ER to the ERGIC or Golgi apparatus.
 - Brefeldin A prevents GTP binding to ARF, and thus inhibits formation of COPI vesicles.
 - Botulinum toxin B acts by cleaving synaptobrevin, inhibiting release of acetylcholine at the neuromuscular junction.
 - Furin converts preproalbumin to proalbumin.
31. Which one of the following types of protein does NOT act as a GTPase?
- ADP ribosylation factor (ARF)
 - Rab proteins
 - N-ethylmaleimide-sensitive factor (NSF)
 - Sar1
 - Ran proteins
32. Select the one FALSE statement:
- Collagen has a triple helical structure, forming a right-hand superhelix.
 - Proline and hydroxyproline confer rigidity on collagen.
 - Collagen contains one or more O-glycosidic linkages.
 - Collagen lacks cross-links.
 - Deficiency of vitamin C impairs the action of prolyl and lysyl hydroxylases.
33. Select the one FALSE statement:
- Elastin contains hydroxyproline, but not hydroxylysine.
 - Elastin contains cross-links formed by desmosines.
 - No genetic diseases due to abnormalities of elastin have as yet been identified.
 - Unlike collagen, there is only one gene encoding elastin.
 - Elastin does not contain any sugar molecules.
34. Select the one FALSE statement:
- Marfan syndrome is due to mutations in the gene encoding fibrillin-1, a major constituent of microfibrils.
 - All subtypes of Ehlers-Danlos syndrome are due to mutations affecting the genes encoding the various types of collagen.
 - Laminin is found in renal glomeruli along with entactin, type IV collagen, and heparin or heparan sulfate.
 - Mutations affecting type IV collagen can cause serious renal disease.
 - Mutations in the collagen *IA1* gene can cause osteogenesis imperfecta.
35. Select the one FALSE statement:
- Most but not all GAGs contain an amino sugar and a uronic acid.
 - All GAGs are sulfated.
 - GAGs are built up by the actions of glycosyltransferases using sugars donated by nucleotide sugars.
 - Glucuronic acid can be converted to iduronic acid by an epimerase.
 - The proteoglycan aggrecan contains hyaluronic acid, keratan sulfate, and chondroitin sulfate.

36. A male infant is failing to thrive and, on examination, is noted to have hepatomegaly and splenomegaly, among other findings. Urinalysis reveals the presence of both dermatan sulfate and heparan sulfate. You suspect the patient has Hurler syndrome. From the following list, select the enzyme that you would wish to have assayed to support your diagnosis:
- A. β -Glucuronidase
 - B. β -Galactosidase
 - C. α -L-Iduronidase
 - D. α -N-Acetylglucosaminidase
 - E. Neuraminidase
37. You see a child in clinic who is well below average height. You note that the child has short limbs, normal trunk size, macrocephaly, and a variety of other skeletal abnormalities. You suspect that the child has achondroplasia. Select from the following list the test that would best confirm your diagnosis:
- A. Measurement of growth hormone
 - B. Assays for enzymes involved in the metabolism of GAGs
 - C. Tests for urinary mucopolysaccharides
 - D. Gene tests for abnormalities of the fibroblast growth factor receptor 3 (FGFR3)
 - E. Gene tests for abnormalities of growth hormone

Exam Questions

Section XI – Special Topics (C)

- Which one of the following statements regarding the blood coagulation pathways is NOT CORRECT?
 - The components of the extrinsic Xase (tenase) complex are factor VIIa, tissue factor, Ca^{2+} , and factor X.
 - The components of the intrinsic Xase (tenase) complex are factors IXa and VIIIa, Ca^{2+} , and factor X.
 - The components of the prothrombinase complex are factors Xa and Va, Ca^{2+} , and factor II (prothrombin).
 - The extrinsic and intrinsic Xase complexes and prothrombinase complex require anionic procoagulant phosphatidylserine on low-density lipoprotein (LDL) for their assembly.
 - Fibrin formed by cleavage of fibrinogen by thrombin is covalently cross-linked by the action of factor XIIIa, which itself is formed by the action of thrombin on factor XIII.
- On which one of the following coagulation factors does a patient taking warfarin for his thrombotic disorder have decreased Glu (γ -carboxyglutamate) residues?
 - Tissue factor
 - Factor XI
 - Factor V
 - Factor II (prothrombin)
 - Fibrinogen
- A 65-year-old man suffers a myocardial infarction and is given tissue plasminogen activator within 6 hours of onset of the thrombosis to achieve which one of the following?
 - Prevent activation of the extrinsic pathway of coagulation
 - Inhibit thrombin
 - Enhance degradation of factors VIIIa and Va
 - Enhance fibrinolysis
 - Inhibit platelet aggregation
- Which one of the following statements regarding platelet activation in hemostasis and thrombosis is NOT CORRECT?
 - Platelets adhere directly to subendothelial collagen via GPIa-IIa and GPIIb/IIIa, while binding of GPIb-IX-V is mediated via von Willebrand factor.
 - The aggregating agent thromboxane A_2 is formed from arachidonic acid liberated from platelet membrane phospholipids by the action of phospholipase A_2 .
 - The aggregating agent ADP is released from the dense granules of activated platelets.
 - The aggregating agent thrombin activates intracellular phospholipase $\text{C}\beta$, which forms the internal effector molecules 1,2-diacylglycerol and 1,4,5-inositol trisphosphate from the membrane phospholipid phosphatidylinositol 4,5-bisphosphate.
 - The ADP receptors, the thromboxane A_2 receptor, the thrombin PAR-1 and PAR-4 receptors, and the fibrinogen GPIIb-IIIa receptor are all examples of G-protein-coupled receptors.
- A 15-year-old adolescent girl presented at clinic with bruises on her lower extremities. Of the following, which is *least likely* to explain the bleeding signs exhibited by this individual?
 - Hemophilia A
 - von Willebrand disease
 - A low platelet count
 - Aspirin ingestion
 - A platelet disorder with absence of storage granules
- Regarding chemical carcinogenesis, select the one FALSE statement:
 - Approximately 80% of human cancers may be due to environmental factors.
 - In general, chemical carcinogens interact noncovalently with DNA.
 - Some chemicals are converted to carcinogens by enzymes, usually cytochrome P450 species.
 - Most ultimate carcinogens are electrophiles and attack nucleophilic groups in DNA.
 - The Ames assay is a useful test for screening chemicals for mutagenicity; however, animal testing is required to show that a chemical is carcinogenic.
- Regarding viral carcinogenesis, select the one FALSE statement:
 - Approximately 15% of human cancers may be caused by viruses.
 - Only RNA viruses are known to be carcinogens.
 - RNA viruses causing or associated with tumors include hepatitis C virus.
 - Retroviruses possess reverse transcriptase, which copies RNA to DNA.
 - Tumor viruses act by deregulating the cell cycle, inhibiting apoptosis, and interfering with normal cell signaling processes.
- Regarding oncogenes and tumor suppressor genes, select the one FALSE statement:
 - Both copies of a tumor suppressor gene must be mutated for its product to lose its activity.
 - Mutation of an oncogene occurs in somatic cells and is not inherited.
 - The product of an oncogene shows a gain of function that signals cell division.
 - RB* and *P53* are tumor suppressor genes; *MYC* and *RAS* are oncogenes.
 - Mutation of one tumor suppressor gene or one oncogene is thought to be sufficient to cause cancer.

9. Regarding growth factors, select the one FALSE statement:
- They include a large number of polypeptides, most of which stimulate cell growth.
 - Growth factors can act in an endocrine, paracrine, or autocrine manner.
 - Certain growth factors, such as TGF- β , can act in a growth inhibitory manner.
 - Some receptors for growth factors have tyrosine kinase activity; mutations of these receptors occur in cancer cells.
 - PDGF stimulates phospholipase A₂, which hydrolyzes PIP₂ to form DAG and IP₃, both of which are second messengers.
10. Regarding the cell cycle, select the one FALSE statement:
- Cells transiting the cell cycle can reside within any of the five phases of the cell cycle (ie, G₁, G₀, S, G₂, and M).
 - Cancer cells usually have a shorter generation time than normal cells and there are fewer of them in G₀ phase.
 - A variety of mutations in cyclins and CDKs have been reported in cancer cells.
 - RB is a cell cycle regulator, where it binds to transcription factor E2F, thus allowing progression of the cell from G₁ to S phase.
 - When damage to DNA occurs, p53 increases in amount and activates transcription of genes that delay transit through the cycle.
11. Regarding chromosomes and genomic instability, select the one FALSE statement:
- Cancer cells may have a mutator phenotype, which means that they have mutations in genes that affect DNA replication and repair, chromosomal segregation, DNA damage surveillance, and apoptosis.
 - Chromosomal instability refers to gain or loss of chromosomes caused by abnormalities of chromosomal segregation during mitosis.
 - Microsatellite instability involves expansion or contraction of microsatellites due to abnormalities of nucleotide excision repair.
 - Aneuploidy (when the chromosomal number of a cell is not a multiple of the haploid number) is a common feature of tumor cells.
 - Abnormalities of chromosome cohesion and of kinetochore-microtubule attachment may contribute to chromosomal instability and aneuploidy.
12. Select the one FALSE statement:
- The activity of telomerase is frequently elevated in cancer cells.
 - A number of cancers have a strong hereditary predisposition or susceptibility; these include Li-Fraumeni syndrome and retinoblastoma.
 - The products of *BRCA1* and *BRCA2* (responsible for hereditary breast cancer types I and II) appear to be involved in DNA repair.
 - Tumor cells usually exhibit a high rate of anaerobic glycolysis; this may be at least partly explained by the presence in many tumor cells of the PK-2 isozyme, which is associated with lesser production of ATP and possibly increased use of metabolites to build up biomass.
 - Dichloroacetate, a compound found to display some anticancer activity, inhibits pyruvate carboxylase, and thus diverts pyruvate away from glycolysis.
13. Select the one FALSE statement:
- Whole-genome and exome sequencing is revealing important new information about the numbers and types of mutations in cancer cells.
 - Abnormalities of epigenetic mechanisms, such as demethylation of cytosine residues, abnormal modification of histones, and aberrant chromatin remodeling are being increasingly detected in cancer cells.
 - Persistence of cancer stem cells (which are often relatively dormant and have active DNA repair systems) may help to explain some of the shortcomings of chemotherapy.
 - Angiogenin is an inhibitor of angiogenesis.
 - Chronic inflammation, possibly via increased production of reactive oxygen species, predisposes to development of certain types of cancer.
14. Regarding apoptosis, select the one FALSE statement:
- Apoptosis can be initiated by the interaction of certain ligands with specific receptors on cell surface.
 - Cell stress and other factors activate the mitochondrial pathway of apoptosis; release of cytochrome P450 into the cytoplasm is an important event in this pathway.
 - A distinct pattern of fragments of DNA is found in apoptotic cells; it is caused by caspase-activated DNase.
 - Caspase 3 digests cell proteins such as lamin, certain cytoskeletal proteins, and various enzymes, leading to cell death.
 - Cancer cells have acquired various mutations that allow them to evade apoptosis, prolonging their existence.
15. Select the one FALSE statement:
- Proteins involved in cell adhesion include cadherins, integrins, and selectins.
 - Decreased amounts of E-cadherin on the surfaces of cancer cells may help account for the decreased adhesiveness shown by tumor cells.
 - Increased activity of GlcNAc transferase V in cancer cells may lead to an altered glycan lattice at the cell surface, perhaps predisposing to their spread.
 - Cancer cells secrete metalloproteinases that degrade proteins in the ECM and facilitate their spread.
 - All tumor cells have the genetic capacity to colonize.
16. The number of enzymes dedicated to repairing hydrolytic, oxidative, and photochemical damage to polynucleotides such as DNA is much greater than the number devoted to repairing damaged proteins. Identify the statement from the list below that is INCONSISTENT with this observation:
- Polynucleotides absorb ultraviolet light more efficiently than do proteins.
 - Proteins contain sulfur, an element that is susceptible to oxidation.
 - In general, proteins turn over more frequently than does DNA.
 - Mutations in a structural gene have the potential to alter the proteins they encode as well as the DNA itself.
 - If left uncorrected, genome mutations will be passed on to succeeding generations.
17. Which of the following is NOT a feature of the mitochondrial
- Calorically restricted diets promote lower and more

- hereditary cancer types I and II appear to be involved in DNA repair.
- D. Tumor cells usually exhibit a high rate of anaerobic glycolysis; this may be at least partly explained by the presence in many tumor cells of the PK-2 isozyme, which is associated with lesser production of ATP and possibly increased use of metabolites to build up biomass.
- E. Dichloroacetate, a compound found to display some anticancer activity, inhibits pyruvate carboxylase, and thus diverts pyruvate away from glycolysis.
- C. In general, proteins turn over more frequently than does DNA.
- D. Mutations in a structural gene have the potential to alter the proteins they encode as well as the DNA itself.
- E. If left uncorrected, genome mutations will be passed on to succeeding generations.

17. Which of the following is NOT a feature of the mitochondrial hypothesis of aging?
- Reactive oxygen species are generated as a by-product by the electron transport chain.
 - Mitochondria lack the capacity to repair damaged DNA.
 - Many of the complexes in the electron transport chain are constructed from a mixture nuclearly encoded and mitochondrially encoded subunits.
 - Damaged mitochondria form protease-resistant aggregates.
 - Damaged mitochondria can trigger apoptosis—programmed cell death.
18. Which of the following is NOT a component of the cell's suite of damage repair and prevention agents?
- Superoxide dismutase
 - Glutathione
 - Isoaspartyl methyltransferase
 - Catalase
 - Caspase 7
19. Select the one of the following statements that describes an aspect of the metabolic theory of aging:
- Elevated levels of plasma glucose promote the formation of cross-linked protein aggregates.
 - Damage from ROS is multiplied by the tendency of oxygen radicals to multiply via chain reactions.
 - Calorically restricted diets promote lower and more efficient metabolic activity.
 - Blood flow to the heart muscle becomes restricted over time due to the cholesterol-induced formation of arterial plaques.
 - Vigorous physical activity correlates with the loss of STEM cells.
20. Select the one of the following statements that is NOT CORRECT:
- Telomeres prevent genetic recombination by capping the ends of linear DNA molecules.
 - Aging genes can be distinguished by their impact on an organism's lifespan.
 - The short lifespan of *Caenorhabditis elegans* renders them an attractive model organism for studying aging.
 - Telomere shortening is a consequence of the discontinuous nature of the process by which the "lagging strand" is synthesized during chromosome replication.
 - Telomerase activity is high in both STEM cells and in many cancer cells.



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Section I – Proteins: Structure & Function

1. B.
2. D.
3. That fermentation required intact cells was disproved by the discovery that a cell-free yeast extract could convert sugar to ethanol and carbon dioxide. This discovery led to the identification of the intermediates, enzymes, and cofactors of fermentation and glycolysis.
4. Fermentation ceased over time, but resumed when inorganic orthophosphate was added. This led to the isolation of phosphorylated intermediates. Other experiments using heated yeast extract led to the discovery of ATP, ADP, and NAD.
5. Preparations used to identify metabolites and enzymes included perfused liver, liver slices, and tissue homogenates fractionated by centrifugation.
6. Radioactive ^{14}C , ^3H , and ^{32}P facilitated the isolation of intermediates of carbohydrate, lipid, nucleotide, and amino acid metabolism and enabled precursor product relationships between intermediates to be tracked.
7. Garrod proposed that alkaptonuria, albinism, cystinuria, and pentosuria resulted from "inborn errors of metabolism" led to the field of biochemical genetics.
8. Regulation of cholesterol biosynthesis illustrates the link between biochemistry and genetics. Cell surface receptors internalize plasma cholesterol, which then regulates cholesterol biosynthesis. Defective receptors result in extreme hypercholesterolemia.
9. Key model organisms include yeast, slime mold, fruit fly, and a small round worm, each with a short generation time and readily mutated.
10. D. Hydrocarbons are water insoluble.
11. A. Of the protein amino acids, only phenylalanine, tyrosine, and tryptophan absorb light at 280 nm.
12. D. When present in solution at a pH equal to their pK_a only half of the molecules of a monofunctional weak acid (eg, ammonium ion or acetic acid) are in the charged state. Maximal mobility will occur either at a pH 3 or more pH units below the pK_a for ammonium ion, or at a pH 3 or more pH units above the pK_a for acetic acid.
13. C. At its pI an amino acid has an equal number of positive and negative charges, but has no *net* overall charge.
14. C. The Edman technique involves successive derivatization and removal of N-terminal residues.
15. Self-association in an aqueous environment as a large droplet minimizes the surface area in contact with water, and hence the number of water molecules whose degrees of rotational freedom are restricted.
16. Strong bases and acids dissociate essentially completely in water, NaOH as Na^+ and OH^- . By contrast, a weak acid such as pyruvic acid dissociates only partially in solution.
17. E. Tandem mass spectrometry can separate complex mixtures of peptides.
18. E. Many proteins undergo posttranslational processing, for example, insulin, which is synthesized as a single polypeptide which subsequent proteolysis converts to two polypeptide chains linked by disulfide bonds.
19. pI is the pH at which a molecule bears no *net* charge. In this example, the pI is a pH midway between the third and fourth pK_a values: $\text{pI} = (6.3 + 7.7)/2 = 7.0$. As pH is adjusted from acidic to basic, net charge will change successively as follows: +3, +2, +1, 0, -1, -2, -3.
20. All of the protein amino acids are *essential* since all are required for protein synthesis, but "nutritionally essential" amino acids (10 for humans) are those which an organism cannot synthesize. Many vitamins are "dietarily essential," although vitamin C is *dietarily essential* only for humans, catfish, and certain other organisms.
21. D. Gene arrays, also termed DNA chips or DNA arrays, contain multiple DNA probes with differing sequences bound at known locations on a solid support. Hybridization of complementary DNA or RNA probes at particular locations provides information about their nucleic acid composition.
22. D. A hydrogen bond interaction involves the residue in fourth place along the helix.
23. E. Prions contain no nucleic acid, just protein. Prion diseases therefore are transmitted by protein without involvement of DNA or RNA.
24. Unlike pK_2 (6.82) of phosphoric acid, the other two dissociating groups of phosphoric acid cannot serve as effective buffers at physiologic pH because they are either completely dissociated or predominantly protonated at pH 7.
25. A: Carboxyl groups (pK_1 through pK_3) and amino groups (pK_4 through pK_5)
B: Minus one
C: Plus 0.5
D: Toward the cathode
26. To act as an effective buffer, a compound should have a pK_a no less than 0.5 pH units removed from the desired pH, and be present in sufficient quantity.
27. Carboxylation of a glutamyl residue forms γ -carboxyglutamate, a potent chelator of Ca^{++} required for blood clotting and clot dissolution. 4-Hydroxyproline and 5-hydroxylysine are present in several structural proteins.
28. (a) Copper is an essential prosthetic group for the amine oxidase that converts lysine to the hydroxylysine that participates in formation of covalent crosslinks that strengthen collagen.
(b) Ascorbic acid is essential for proline hydroxylase to convert proline to the hydroxyproline, which provides interchain hydrogen bonds that stabilize the collagen triple helix.
29. Signal sequences target proteins to specific subcellular locations in the cell, or for secretion.

Section II – Enzymes: Kinetics, Mechanism, Regulation, & Role of Transition Metals

1. Carbonic anhydrase catalyzes the hydration of carbon dioxide to form carbonic acid. A portion of this weak acid, in turn, dissociates to produce bicarbonate and a proton. As the concentration of carbon dioxide falls, carbonic acid is broken down to form carbon dioxide and water. To compensate for the loss of

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carbonic acid, bicarbonate and protons recombine to restore equilibrium, leading to a net drop in $[H^+]$ and a rise in pH.

- | | | | |
|-------|--------|--------|--------|
| 2. D. | 7. B. | 12. B. | 17. B. |
| 3. E. | 8. C. | 13. B. | 18. D. |
| 4. B. | 9. A. | 14. C. | 19. A. |
| 5. A. | 10. D. | 15. D. | |
| 6. E. | 11. E. | 16. A. | |

Section III – Bioenergetics

1. A. A reaction with a negative ΔG is exergonic, it proceeds spontaneously and free energy is released.
2. E. In an exergonic reaction ΔG is negative and in an endergonic reaction it is positive. When ΔG is zero, the reaction is at equilibrium.
3. B. When the reactants are present in concentrations of 1.0 mol/L, ΔG° is the standard free-energy change. For biochemical reaction, the pH (7.0) is also defined and this is $\Delta G'^\circ$.
4. D. ATP contains two high-energy phosphate bonds and is needed to drive endergonic reactions. It is not stored in the body and in the presence of uncouplers its synthesis is blocked.
5. A. Reduced cytochrome c is oxidized by cytochrome c oxidase (complex IV of the respiratory chain), with the concomitant reduction of molecular oxygen to two molecules of water.
6. E. Cytochrome oxidase is not a dehydrogenase, although all other cytochromes are classed as such.
7. B. Although Cytochromes p450 are located mainly in the endoplasmic reticulum, they are found in mitochondria in some tissues.
8. D. Oxidation of one molecule of NADH via the respiratory chain generates 2.5 molecules of ATP in total. One is formed via complex I, 1 via complex II and 0.5 via complex IV.
9. C. 1.5 molecules of ATP are formed in total as $FADH_2$ is oxidized, 1 via complex II and 0.5 via complex IV.
10. E. Oligomycin blocks oxidation and ATP synthesis as it prevents the flow of electrons back into the mitochondrial matrix through ATP synthase.
11. A. Uncouplers allow electrons to reenter the mitochondrial matrix without passing through ATP synthase.
12. E. In the presence of an uncoupler, the energy released as electrons flow into the mitochondrial matrix is not captured as ATP and is dissipated as heat.
13. C. Thermogenin is a physiological uncoupler found in brown adipose tissue. Its function is to generate body heat.
14. D. Three ATP molecules are generated for each revolution of the ATP synthase molecule.
15. B. The electrochemical potential difference across the inner mitochondrial membrane caused by electron transport must be negative on the matrix side so that protons are forced to reenter via the ATP synthase to discharge the gradient.

Section IV – Metabolism of Carbohydrates

- | | | | |
|-------|--------|--------|--------|
| 1. B. | 8. B. | 15. D. | 22. A. |
| 2. B. | 9. B. | 16. E. | 23. B. |
| 3. A. | 10. E. | 17. E. | 24. C. |
| 4. D. | 11. C. | 18. C. | 25. D. |
| 5. C. | 12. D. | 19. C. | 26. E. |
| 6. C. | 13. D. | 20. C. | 27. A. |
| 7. E. | 14. D. | 21. D. | 28. B. |

Section V – Metabolism of Lipids

1. D.
2. D.
3. A. Gangliosides are derived from glucosylceramide.
4. C. A, B, D, and E are classed as preventive antioxidants as they act by reducing the rate of chain initiation.
5. D.
6. B.
7. D. Long chain fatty acids are activated by coupling to CoA, but fatty acyl CoA cannot cross the inner mitochondrial membrane. After transfer of the acyl group from CoA to carnitine by carnitine palmitoyl transferase (CPT)-I, acylcarnitine is carried across by carnitine-acylcarnitine translocase in exchange for a carnitine. Inside the matrix, CPT-II transfers the acyl group back to CoA and carnitine is taken back into the intermembrane space by the translocase enzyme.
8. E. The breakdown of palmitic acid (C16) requires 7 cycles of β -oxidation each producing 1 $FADH_2$ and 1 NADH molecule and results in the formation of eight 2C acetyl CoA molecules.
9. B. When the action of carnitine palmitoyl transferase-I is inhibited by malonyl CoA, fatty acyl groups are unable to enter the matrix of the mitochondria where their breakdown by β -oxidation takes place.
10. C. Humans (and most mammals) do not possess enzymes able to introduce a double bond into fatty acids beyond $\Delta 9$.
11. D. Inhibition of the tricarboxylic acid transporter causes levels of citrate in the cytosol to decrease and favors inactivation of the enzyme.
12. A.
13. C.
14. E.
15. E. Glucagon is released when blood glucose levels are low. In this situation, fatty acids are broken down for energy and fatty acid synthesis is inhibited.
16. E. Glucagon, ACTH, epinephrine and vasopressin promote activation of the enzyme.
17. B.
18. D.
19. A. Chylomicrons are triacylglycerol-rich lipoproteins synthesized in the intestinal mucosa using fat from the diet and secreted into lymph.
20. E. VLDL is synthesized and secreted by the liver, and adipose tissue and muscle take up the fatty acids released by the action of lipoprotein lipase.
21. D. Very low density lipoprotein secreted by the liver is converted to intermediate density lipoprotein and then to low density lipoprotein (LDL) by the action of lipases and the transfer of cholesterol and proteins from high density lipoprotein. LDL delivers cholesterol to extrahepatic tissues and is also cleared by the liver.
22. A. Chylomicrons are synthesized in the intestine and secreted into lymph after a fat meal.
23. E. Chylomicrons and their remnants are cleared from the circulation rapidly after a meal, and the secretion of very low density lipoprotein by the liver then increases. Ketone bodies and non-esterified fatty acids are elevated in the fasting state.
24. C. When cholesteryl ester is transferred from HDL to other lipoproteins by the action of CETP it is ultimately delivered to the liver in VLDL, IDL, or LDL.

25. D. Chylomicrons are metabolized by lipoprotein lipase when bound to the surface of endothelial cells. This process releases

14. Since glutamate dehydrogenase plays multiple central roles in metabolism, its complete absence would unquestionably be

25. D. Chylomicrons are metabolized by lipoprotein lipase when bound to the surface of endothelial cells. This process releases fatty acids from triacylglycerol which are then taken up by the tissues. The resulting smaller, cholesterol-enriched chylomicron remnant particles are released into the circulation and cleared by the liver.
26. C. Cholesterol is synthesized in the endoplasmic reticulum from acetyl CoA. The rate-limiting step is the formation of mevalonate from 3-hydroxy 3-methylglutaryl-CoA by HMG CoA reductase and lanosterol is the first cyclic intermediate.
27. C.
28. C. Secondary bile acids are produced by the modification of primary bile acids in the intestine.
29. B. If the LDL receptor is defective, LDL is not cleared from the blood, causing severe hypercholesterolemia.
30. A. PCSK9 regulates the re-cycling of LDL receptors to the cell surface after endocytosis has taken place. Inhibition of PCSK9 activity, therefore, increases the number of LDL receptor molecules on the cell surface, leading to an increased rate of clearance and lower blood cholesterol levels.
14. Since glutamate dehydrogenase plays multiple central roles in metabolism, its complete absence would unquestionably be fatal.
15. E. Albumin is not a hemoprotein. In cases of hemolytic anemia, albumin can bind some metheme, but unlike the other proteins listed, albumin is not a hemoprotein.
16. A. Acute intermittent porphyria is due to mutations in the gene for uroporphyrin I synthase.
17. A. Bilirubin is a linear tetrapyrrole.
18. D. The severe jaundice, upper abdominal pain, and weight loss plus the lab results indicating an obstructive type of jaundice are consistent with cancer of the pancreas.
19. The assay takes advantage of the different water solubility of unconjugated and conjugated bilirubin. Two assays are conducted, one in the absence and a second in the presence of an organic solvent, typically methanol. The highly polar glucuronic acid groups of conjugated bilirubin convey water solubility that insures that it will react with the colorimetric reagent even in the absence of any added organic solvent. Data from an assay conducted in the absence of added methanol, termed "direct bilirubin," is bilirubin glucuronide. A second assay conducted in the presence of added methanol measures total bilirubin, ie, both conjugated and unconjugated bilirubin. The difference between total bilirubin and direct bilirubin, reported as "indirect bilirubin," is unconjugated bilirubin.
20. The biosynthesis of heme from succinyl-CoA and glycine occurs only when the availability of free iron signals the potential for synthesis of heme. Regulation targets the first enzyme of the pathway, δ -aminolevulinic synthase (ALA synthase) rather than a subsequent reaction. This conserves energy by avoiding wasting a coenzyme A thioester.

Section VI – Metabolism of Proteins & Amino Acids

1. D. Phenylalanine hydroxylase catalyzes a functionally irreversible reaction, and thus cannot convert tyrosine to phenylalanine.
2. E. Histamine is a catabolite, not a precursor, of histidine.
3. B. The insertion of selenocysteine into a peptide occurs during, not subsequent to translation.
4. C. Pyridoxal-dependent transamination is the first reaction in degradation of all the common amino acids except threonine, lysine, proline, and hydroxyproline.
5. B. Glutamine.
6. C. The carbon skeleton of alanine contributes the most to hepatic gluconeogenesis.
7. B. ATP and ubiquitin participate in the degradation of nonmembrane-associated proteins and proteins with short half-lives.
8. C. Due to the failure to incorporate NH_4^+ into urea, clinical signs of metabolic disorders of the urea cycle include alkalosis, not acidosis.
9. E. Cytosolic fumarase and cytosolic malate dehydrogenase convert fumarate to oxaloacetate following a cytosolic reaction of the urea cycle. The mitochondrial fumarase and malate dehydrogenase function in the TCA cycle, not urea biosynthesis.
10. A. Serine, not threonine, provides the thioethanol moiety of coenzyme A.
11. E. Decarboxylation of glutamate, not glutamine forms GABA.
12. 5-Hydroxylysine and γ -carboxyglutamate represent examples of posttranslational modification of peptidyl lysyl and peptidyl glutamyl residues, respectively. By contrast, selenocysteine is incorporated into proteins cotranslationally, in the same way as the other 20 common protein amino acids. The process is complex, and involves the unusual tRNA termed tRNA^{sec}.
13. Biosynthesis of the amino acids that are dietarily essential for humans requires multiple reactions. Since human diets typically contain adequate amounts of these amino acids, loss of the genes that can encode these "unnecessary" enzymes and the lack of need to expend the energy required to copy them provide an evolutionary advantage.

Section VII – Structure, Function, & Replication of Informational Macromolecules

1. D. β,γ -Methylene and β,γ -imino purine pyrimidine triphosphates do not readily release the terminal phosphate by hydrolysis or by phosphoryl group transfer.
2. D.
3. E. Pseudouridine is excreted unchanged in human urine. Its presence there is not indicative of pathology.
4. A. Metabolic disorders are infrequently associated with defects in pyrimidine catabolism, which forms water-soluble products.
5. B. 21. C. 37. C. 53. D.
6. D. 22. A. 38. E. 54. A.
7. B. 23. C. 39. D. 55. E.
8. C. 24. A. 40. D. 56. A.
9. C. 25. E. 41. B. 57. E.
10. D. 26. B. 42. A. 58. C.
11. E. 27. A. 43. A. 59. A.
12. B. 28. E. 44. E. 60. D.
13. D. 29. C. 45. C. 61. D.
14. D. 30. A. 46. A. 62. E.
15. E. 31. A. 47. C. 63. A.
16. A. 32. C. 48. D. 64. C.
17. C. 33. D. 49. C. 65. C.
18. B. 34. E. 50. B. 66. E.
19. D. 35. C. 51. E. 67. D.
20. B. 36. B. 52. C.

Section VIII – Biochemistry of Extracellular & Intracellular Communication

1. B. Glycolipids are located on the outer leaflet.
2. A. Alpha-helices are major constituents of membrane proteins.
3. E. Insulin also increases glucose uptake in muscle.
4. A. Its action maintains the high intracellular concentration of potassium compared with sodium.
5. D.
6. B.
7. C.
8. B.
9. D.
10. A.
11. E.
12. B.
13. D.
14. E.
15. B.
16. C.
17. A.
18. C.
19. A.
20. B.
21. D.
22. A.
23. D.
24. D.
25. B.
26. A.

Section IX – Special Topics (A)

- | | | | |
|--------|--------|--------|--------|
| 1. A. | 16. A. | 31. E. | 46. B. |
| 2. E. | 17. B. | 32. A. | 47. B. |
| 3. C. | 18. C. | 33. B. | 48. B. |
| 4. D. | 19. E. | 34. A. | 49. B. |
| 5. E. | 20. D. | 35. B. | 50. C. |
| 6. D. | 21. E. | 36. C. | 51. D. |
| 7. C. | 22. A. | 37. D. | 52. C. |
| 8. B. | 23. C. | 38. E. | 53. A. |
| 9. D. | 24. C. | 39. E. | 54. A. |
| 10. E. | 25. A. | 40. A. | 55. A. |
| 11. C. | 26. E. | 41. D. | 56. E. |
| 12. B. | 27. A. | 42. C. | 57. A. |
| 13. C. | 28. A. | 43. B. | |
| 14. D. | 29. A. | 44. E. | |
| 15. B. | 30. C. | 45. C. | |

Section X – Special Topics (B)

1. Within the body, hydrolysis of nitroglycerin releases nitrate ions that can be reduced by mitochondrial aldehyde dehydrogenase to generate nitric oxide (NO), a potent vasodilator.
2. The contractile cycle of cardiac muscle is controlled by oscillations in the level of cytosolic Ca^{2+} . If the reuptake of Ca^{2+} by the sarcoplasmic reticulum is slowed sufficiently by a deficiency SERCA2a activity, cardiac myocytes will be unable to clear this second messenger from their cytoplasm prior to the onset of the next cycle of excitation. The persistence of high basal levels of cytosolic Ca^{2+} will lead to both a reduction in the amplitude of the contractile cycle and the progressive uncoupling of the excitation-contraction cycle.
3. A.
4. A.
5. B.
6. B.
7. E.
8. D.
9. E.
10. Haptoglobin binds extracorporeal hemoglobin, forming a complex that is too large to pass through the glomerulus into kidney tubules.
11. The production of new antibodies with unique antigen-binding properties is reliant on the recombination and mutation of the DNA encoding the hypervariable regions of the immunoglobulin heavy and light chains. Cytidine deaminase introduces genetic mutations by catalyzing the hydrolysis of cytosine bases present in DNA to uracil.
12. B.
13. C.
14. C.
15. B.
16. E.
17. E.

18. Erythrocytes deficient in glucose-6-phosphate are rendered extremely vulnerable to destruction by reactive oxygen species resulting from a lack of reduced glutathione, an important agent for protecting against oxidative stress. This is a consequence of their reliance on this enzyme to generate a plentiful supply of the NADPH used by glutathione reductase.
19. E.
20. C.
21. D.
22. A.
23. D.
24. D.
25. B.
26. A.
27. E. Importins are involved in the import of proteins into the nucleus.
28. B. Some mammalian proteins are known to be translocated posttranslationally.
29. C. Ubiquitin tags proteins for degradation by proteasomes.
30. E. Fibrinogen converts proalbumin to albumin.
31. C. NSF is an ATPase.
32. D. Cross-links are an important feature of collagen structure.
33. C. Deletions in the elastin gene have been identified as responsible for many cases of Williams-Beuren syndrome.
34. B. Ehlers-Danlos syndrome subtypes kyphoscoliosis and dermatosparaxis are caused by defects in noncollagen genes.
35. B. Hyaluronic acid (hyaluronan) is not sulfated.
36. C. Hurler syndrome is caused by a deficiency of α -L-iduronidase.
37. D. Achondroplasia is caused by mutations in the FGFR3 gene.

Section XI – Special Topics (C)

1. D.
2. D. Of the listed proteins, only factor II is a vitamin K-dependent coagulation factor.
3. D.
4. E. GPIIb-IIIa (integrin α IIb β 3) is not a G protein-coupled receptor.
5. A. Hemophilia A, being an X-chromosome-linked disease, is a very unlikely to occur in a female.
6. B. Most chemical carcinogens interact covalently with DNA.
7. B. Certain DNA viruses are also known to be carcinogenic.
8. E. Mutations in approximately 5 to 6 of these two types of cancer promoting or suppressor genes are thought to be necessary for carcinogenesis.
9. E. PDGF stimulates phospholipase C, not phospholipase A.
10. D. Binding of RB to E2F blocks progression of the cell from G_1 to S phase.
11. C. Microsatellite instability is caused by abnormalities of mismatch repair.
12. E. Dichloroacetate inhibits pyruvate dehydrogenase kinase.
13. D. Angiogenin is an inhibitor of angiogenesis.
14. B. Cytochrome C is released from mitochondria.
15. E. Only about 1 in 10,000 cancer may have the capacity to colonize.
16. B.
17. D.
18. D.
19. C.
20. B.